RWANDA KTRN BSS PROJECT

INTERFACE DEFINITION

<EXTERNAL INTERFACE>

DOC. ID	
VERSION	2.1
FIRST CREATED	2018.04.12
LAST MODIFIED	2026.08.19

[Version Information]

Project	Rwanda oRn BSS Project				
Doc. Name	Interface Definition	Version	2.1	Date	2026/09/01
Config. Mng. ID		Team		Writer	QiYanming

[Revision History]

VERSION	REVISED DATE	REVISION CONTENTS	REVISOR
1.0	2014.06.18	THIS REPORT'S DRAFT WAS CREATED.	KIM, JS
1.1	2014.07.23	THE REQUEST ITEM HAS BEEN CHANGED TO “IMSI” TO “MSISDN”. THE INFORMATION HAS BEEN SUPPLEMENTED.	KIM, JS
1.2	2014.07.29	THE DOMAIN IS CHANGED TO “WWW.4GLTE.RW”.	KIM, JS
1.3	2018.02.14	ADD PRODUCT LIST QUERY INTERFACE	KIM, GC
1.4	2018.03.29	SIMPLE ACTIVATION	KIM, GC
1.5	2018.04.03	LOGIN & PURCHASE API CHANGED	KIM, GC
1.6	2018.04.12	RESPONSE CHANGED FOR HPOP101, HPOP301 ZIPCODE / TAX DELETED FOR HPOP501	KIM, GC
1.7	2018.04.19	NID DELETE ON SEARCH BALANCE NID DELETE ON PURCHASE BUNDLE NO LOGIN ON USER APP ADD ESICIA	KIM, GC
1.8	2018.04.25	SO_ID MANDATORY, TID ADD SO_NM ADDED ON BALANCE / USAGE HPOP402 ADDED	KIM, GC
1.9	2023.04.25	REVISE RESPONSE MESSAGE OF ‘HP00001 HPOP008 SEARCH USAGE BALANCE’ TO SUPPORT VOLTE	WC
2.0	2024.04.17	REVISE REQUEST MESSAGE OF ‘HP00001 HPOP008 SEARCH USAGE BALANCE’ TO SET ISP_CD FIELD AS OPTIONAL	WC
2.1	2026.08.19	REVISE REQUEST MESSAGE OF ‘HP00005 HPOP403 YOU CAN CHECK THE BUNDLE PRODUCT PURCHASE OR NOT	WC

TABLE OF CONTENTS

1. Interface Overview 6

1.1. Interface summary 6

2. BUSINESS 7

2.1. Searching Service 7

2.1.1. Searching available bundles (SVCCODE: HP00005, OPCODE: HPOP102) 7

2.1.2. Searching Usage Balance (SVCCODE: HP00001, OPCODE: HPOP008) 9

2.1.3. Searching bundles category list (SVCCODE: HP00005, OPCODE: HPOP103) 12

2.1.4. Searching available bundles on category (SVCCODE: HP00005, OPCODE: HPOP104) 15

2.2. Auth Service 19

2.2.1. Request auth login (SVCCODE: HP00005, OPCODE: HPOP201) 19

2.3. Activation Service 22

2.3.1. Request searching price plan (SVCCODE: HP00005, OPCODE: HPOP301) 22

2.3.2. Request for Simple activation (SVCCODE: HP00005, OPCODE: HPOP302) 25

2.3.3. Searching available bundles (SVCCODE: HP00001, OPCODE: HPOP005) 30

2.3.4. Request for NIDA getCitizen API (SVCCODE: HP00005, OPCODE: HPOP303) 34

2.3.5. Request for Get company information with TIN from RRA (SVCCODE: HP00005, OPCODE: HPOP305) 37

2.3.6. Searching Subscriber Count (SVCCODE: HP00005, OPCODE: HPOP310) 39

2.3.7. Get OTP Code (SVCCODE: HP00005, OPCODE: HPOP311) 42

2.3.8. Verifying Fingerprint (SVCCODE: HP00005, OPCODE: HPOP312) 43

2.3.9. Encrypt/Decrypt Sensitive PWD/CERT (SVCCODE: HP00005, OPCODE: HPOP313) 46

2.4. Purchase Product Service 49

2.4.1. Request Bundle Purchase (SVCCODE: HP00005, OPCODE: HPOP401) 49

2.4.2. Request E-Wallet Amount (SVCCODE: HP00005, OPCODE: HPOP402) 52

2.4.3. Request Bundle Purchase Checking (SVCCODE: HP00005, OPCODE: HPOP403) 54

2.4.4. Request Bundle Purchase (SVCCODE: HP00002, OPCODE: HPOP004) 56

2.5. Subscribe Customer Info 59

2.5.1. Subscribe Customer Info (SVCCODE:HP00005 , OPCODE:HPOP501) 59

2.5.2. Searching Customer Info (SVCCODE : HP00005, OPCODE : HPOP502) 61

2.6. Searching MSISDN 64

2.6.1. Checking MSISDN status (SVCCODE: HP00005, OPCODE: HPOP601) 64

2.6.2. Recommend 5 MSISDNS (SVCCODE: HP00005, OPCODE: HPOP602) 66

2.6.3. Searching MSISDN (SVCCODE: HP00001, OPCODE: HPOP003) 68

2.6.4. Searching Balance (SVCCODE: HP00005, OPCODE: HPOP101) 70

2.6.5. Searching Currency Balance (SVCCODE: HP00001, OPCODE: HPOP001) 72
2.6.6. Balance Transfer (SVCCODE: HP00002, OPCODE: HPOP005) 75
2.6.7. Request for recharge (SVCCODE: HP00002, OPCODE: HPOP001) 78
2.6.8. Cancel for recharging (SVCCODE: HP00002, OPCODE: HPOP002) 81
2.7. KTRN Simreg API workflow (SVCCODE: HP00005) 83
2.7.1. staff login (API 201) 83
2.7.2. create customer (API 312 + API 501) 84
2.7.3. New connection (API 313 + API 310 + API 312 + API 311 + API 302) 85

1. INTERFACE OVERVIEW

SYSTEM FOR INTERWORKING PROTOCOL IS DEFINED AS FOLLOWS. AND IT PROVIDES A SERVICE BASED ON THE CONVENTIONS DEFINED.

1.1. Interface summary

FOR DATA LINKAGE OF VARIOUS BUSINESS, THE ORN BSS SYSTEM PROVIDES THE INFORMATION SERVICE OF THE INTERWORKING.

COMMON MESSAGE

MESSAGE CODE	MESSAGE DESCRIPTION
1001	IT HAS BEEN SUCCESSFULLY PROCESSED.
-4001	ACTIVATOR IDENTIFIER CAN NOT BE FOUND FROM THE GATEWAY.
-4002	ERROR OF DATA TYPE THAT CAN NOT BE CONVERTED AT THE GATEWAY HAS OCCURRED.
-4003	AN ERROR HAS OCCURRED IN THE BUSINESS PROCESS.
-6001	XXXX IS MISSING. (WHEN THE PARAMETER IS MISSING)
-6002	UNDEFINED VALUE/INCORRECT VALUE(XXXX) WAS ENTERED.
-7001	**WORK THAT CAN NOT BE PROCESSED HAS OCCURRED.(XX….XXXX)**
-7002	DATA IS NOT FOUND..

2. BUSINESS

2.1. Searching Service

2.1.1. Searching available bundles (SVCCODE: HP00005, OPCODE: HPOP102)

■ SUMMARY

YOU CAN BUNDLE LIST THAT AVAILABLE FOR PURCHASE AT THE NUMBER MATCH ON MSISDN.

■ TELECOMMUNICATION PROTOCOL FOR REQUEST

REQUEST URI	http://www.4glte.rw/voucher/api/external_access.jsp		
PROTOCOL TYPE	HTTP	METHOD	POST
PORT	80	CONTENTS TYPE	APPLICATION/XML
I/F CYCLE	EVERY REQUEST TIME		

■ REQUEST ITEM / SCHEMA

SEND PARAMETERS (**NAME, VALUE**) : (REQ_XML, XML COMPRISING THE FOLLOWING ELEMENTS)

ELEMENT	M/O	FORMAT	DESCRIPTION
REQUEST	**M**		**REQUEST ROOT**
HEADER	**M**		**REQUEST HEADER**
SVCCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
BODY	**M**		**REQUEST PARAMETER BODY**
MSISDN	M	STRING(10)	MSISDN (CUSTMER PHONE NUMBER)

■ RESPONSE ITEM / SCHEMA

ELEMENT	M/O	FORMAT	DESCRIPTION
RESPONSE	**M**		**RESPONSE ROOT**
HEADER	**M**		**RESPONSE HEADER**
SVCCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
MSGCODE	O	STRING(8)	MESSAGE CODE
MESSAGE	O	STRING(32)	MESSAGE DESCRIPTION
BODY	**M**		**RESPONSE PARAMETER BODY**
PROD_LIST	M	STRING(2048)	THE PRODUCT LIST ORDER IS MADE UP OF THE PRODUCT CODE, THE PRODUCT NAME, AND THE PRODUCT PRICE, AND THE DELIMITER OF THE PRODUCT IS " | "
SO_ID	M	STRING(3)	RETAILER ID

■ MESSAGE

MESSAGE CODE	MESSAGE DESCRIPTION
-5002	MSISDN'S BUSINESS NO. AND TRANSFER XML'S BUSINESS NO. DO NOT MATCH.
-8001	THERE IS NO MATCHING INFORMATION FOR MSISDN OR DISTINGUISH_ID.
-8002	VIEWED CUSTOMERS ARE NOT ELIGIBLE FOR PREPAYMENT.

EXAMPLE :

YOU SHOULD SEND THE PARAMETER THAT IS SET TO THE STRING IS CONTAINING THE FOLLOWING REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<REQUEST>
 <HEADER>
   <SVCODE>HP00005</SVCODE>
   <OPCODE>HPOP102</OPCODE>
   <CHCODE>A</CHCODE>
 </HEADER>
 <BODY>
      <MSISDN>0771000050</MSISDN>
</BODY>
</REQUEST>
```

- **RESPONSE XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<RESPONSE>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
        <OPCODE>HPOP102</OPCODE>
        <CHCODE>A</CHCODE>
        <MSGCODE>XXXX</MSGCODE>
        <MESSAGE>….</MESSAGE>
    </HEADER>
    <BODY>
        <PROD_LIST>
            PD00000574,ORN 500MB BUNDLE(PRE),2300   |
            PD00000575,ORN 10GB BUNDLE(PRE),39600
        </PROD_LIST>
    </BODY>
</RESPONSE>
```

2.1.2. Searching Usage Balance (SVCCODE: HP00001, OPCODE: HPOP008)

■ SUMMARY

You can be searching usage of package

■ TELECOMMUNICATION PROTOCOL FOR REQUEST

REQUEST URI	http://www.4glte.rw/voucher/api/external_access.jsp

PROTOCOL TYPE	HTTP	METHOD	POST
PORT	80	CONTENTS TYPE	APPLICATION/XML
I/F CYCLE	EVERY REQUEST TIME		

■ REQUEST ITEM / SCHEMA

SEND PARAMETERS (**NAME, VALUE**) : (REQ_XML, XML COMPRISING THE FOLLOWING ELEMENTS)

ELEMENT	M/O	FORMAT	DESCRIPTION
REQUEST	**M**		**REQUEST ROOT**
HEADER	**M**		**REQUEST HEADER**
SVCCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
BODY	**M**		**REQUEST PARAMETER BODY**
MSISDN	M	STRING(10)	MSISDN (CUSTMER PHONE NUMBER)
ISP_CD	O	STRING(3)	BUSINESS NO (DEFINED BY ORN)

RESPONSE ITEM / SCHEMA

ELEMENT		M/O	FORMAT	DESCRIPTION
RESPONSE		**M**		**RESPONSE ROOT**
HEADER		**M**		**RESPONSE HEADER**
	SVCCODE	M	STRING(8)	SERVICE CODE
	OPCODE	M	STRING(8)	OPERATION CODE
	CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
	MSGCODE	O	STRING(8)	MESSAGE CODE
	MESSAGE	O	STRING(32)	MESSAGE DESCRIPTION
BODY		**M**		**RESPONSE PARAMETER BODY**
	PURCHASED_LIST	M	STRING(2048)	PRODUCT CODE, PRODUCT NAME, PRODUCT FLAG(V-BUNDLE,B-BASE),PACKAGE PRICE,EFF_DATETIME,EXP_DATETIME, {ALL_BAL_LIST} | NODE: THE 'ALL_BAL_LIST' ARRAY HAVE 'BAL_CATG,BAL_TYPE, ALLOWANCE,USAGE,REMAINING,BAL_EFF_DATETIME,BAL_EXP_DATETIME' OBJECTS. 'BAL_CATG' OBJECT CAN BE 'DATA','SMS','VOICE' VALUES AND UNIT OF 'VOICE' WILL BE SECONDS; 'BAL_TYPE' WILL BE INDEX OF EACH KIND BALANCE, YOU CAN GET BALANCE TYPE LIST FROM KT.

EXAMPLE :

YOU SHOULD SEND THE PARAMETER THAT IS SET TO THE STRING IS CONTAINING THE FOLLOWING REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<REQUEST>
 <HEADER>
   <SVCODE>HP00005</SVCODE>
   <OPCODE>HPOP102</OPCODE>
   <CHCODE>A</CHCODE>
 </HEADER>
```

```
<BODY>
    <MSISDN>0771000050</MSISDN>
    <ISP_CD>62</ISP_CD>
</BODY>
</REQUEST>
```

- **RESPONSE XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<RESPONSE>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
        <OPCODE>HPOP102</OPCODE>
        <CHCODE>A</CHCODE>
        <MSGCODE>XXXX</MSGCODE>
        <MESSAGE>….</MESSAGE>
    </HEADER>
    <BODY>
        <PURCHASED_LIST>
            PI00070600,INFIX PROMO 10GB BUNDLE(PRE),V,1000,2023-12-12 11:20:35,2024-01-11 11:20:35,{DATA,1,10240,10240,0,2023-12-12 11:20:35,2024-01-11 11:20:35},{DATA,2,3072,3072,0,2023-12-12 11:20:35,2023-12-13 11:20:35}|
        PI00070601,INFIX PROMO MIXED BUNDLE(PRE),V,2000,2023-12-10 11:20:35,2024-01-09 11:20:35,{VOICE,3,3600,3000,600,2023-12-10 11:20:35,2024-01-09 11:20:35}|
        PI00070602,INFIX PROMO MIXED BUNDLE(PRE),V,3000,2023-12-12 11:20:35,2024-01-11 11:20:35,{DATA,1,10240,10240,0,2023-12-12 11:20:35,2024-01-11 11:20:35},{VOICE,3,3600,2000,1600,2023-12-12 11:20:35,2024-01-11 11:20:35},{SMS,4,100,30,70,2023-12-12 11:20:35,2024-01-11 11:20:35}
        </PURCHASED_LIST>
    </BODY>
</RESPONSE>
```

2.1.3. Searching bundles category list (SVCCODE: HP00005, OPCODE: HPOP103)

- SUMMARY

  YOU CAN BUNDLE CATEGORY LIST

- TELECOMMUNICATION PROTOCOL FOR REQUEST

REQUEST URI	http://www.4glte.rw/voucher/api/external_access.jsp		
PROTOCOL TYPE	HTTP	METHOD	POST
PORT	80	CONTENTS	APPLICATION/XML

		TYPE	
I/F CYCLE	EVERY REQUEST TIME		

■ REQUEST ITEM / SCHEMA

SEND PARAMETERS (**NAME, VALUE**) : (REQ_XML, XML COMPRISING THE FOLLOWING ELEMENTS)

ELEMENT	M/O	FORMAT	DESCRIPTION
REQUEST	**M**		**REQUEST ROOT**
HEADER	**M**		**REQUEST HEADER**
SVCCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
BODY	**M**		**REQUEST PARAMETER BODY**

RESPONSE ITEM / SCHEMA

ELEMENT	M/O	FORMAT	DESCRIPTION
RESPONSE	**M**		**RESPONSE ROOT**
HEADER	**M**		**RESPONSE HEADER**
SVCCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
MSGCODE	O	STRING(8)	MESSAGE CODE
MESSAGE	O	STRING(32)	MESSAGE DESCRIPTION
BODY	**M**		**RESPONSE PARAMETER BODY**
CATEGORY	M	STRING(2048)	CATEGORY LIST
CATEGORY_CODE	M	STRING(32)	CATEGODY CODE
CATEGORY_PCODE	M	STRING(32)	PARENT CATEGORY CODE
CATEGORY_NAME	M	STRING(32)	CATEGORY NAME
LEVEL	M	STRING(32)	CATEGORY LEVEL (TOP CATEGORY LEVEL : 1)
SUB_CATEGORY	M	STRING(32)	SUB CATEGORY

MESSAGE

MESSAGE CODE	MESSAGE DESCRIPTION
-5002	MSISDN'S BUSINESS NO. AND TRANSFER XML'S BUSINESS NO. DO NOT MATCH.
-8001	THERE IS NO MATCHING INFORMATION FOR MSISDN OR DISTINGUISH_ID.
-8002	VIEWED CUSTOMERS ARE NOT ELIGIBLE FOR PREPAYMENT.

EXAMPLE :

YOU SHOULD SEND THE PARAMETER THAT IS SET TO THE STRING IS CONTAINING THE FOLLOWING REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<REQUEST>
 <HEADER>
```

```
        <SVCODE>HP00005</SVCODE>
        <OPCODE>HPOP103</OPCODE>
        <CHCODE>A</CHCODE>
      </HEADER>
      <BODY>
     </BODY>
     </REQUEST>
```

- **RESPONSE XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<RESPONSE>
  <HEADER>
    <SVCODE>HP00005</SVCODE>
    <OPCODE>HPOP103</OPCODE>
    <CHCODE>A</CHCODE>
    <MSGCODE>1001</MSGCODE>
    <MESSAGE>IT HAS BEEN SUCCESSFULLY PROCESSED. PLEASE PRESS THE CHECK BALANCE BUTTO
N TO CONFIRM THE RESULTS.
    </MESSAGE>
  </HEADER>
  <BODY>
    <CATEGORY>
      <CATEGORY_CODE>C</CATEGORY_CODE>
      <CATEGORY_PCODE>0</CATEGORY_PCODE>
      <CATEGORY_FULLNAME>FIXED PRDT</CATEGORY_FULLNAME>
      <SUB_CATEGORY>
        <CATEGORY_CODE>C01</CATEGORY_CODE>
        <CATEGORY_PCODE>C</CATEGORY_PCODE>
        <CATEGORY_FULLNAME>FIXED PRDT - N/A</CATEGORY_FULLNAME>
         <LEVEL>2</LEVEL>
        <CATEGORY_NAME>N/A</CATEGORY_NAME>
      </SUB_CATEGORY>
        …
      <LEVEL>1</LEVEL>
      <CATEGORY_NAME>FIXED PRDT</CATEGORY_NAME>
    </CATEGORY>
        …
  </BODY>
</RESPONSE>
```

2.1.4. Searching available bundles on category (SVCCODE: HP00005, OPCODE: HPOP104)

- SUMMARY

  YOU CAN BUNDLE LIST THAT AVAILABLE FOR PURCHASE AT THE NUMBER MATCH ON MSISDN ON CATEGORY.

- TELECOMMUNICATION PROTOCOL FOR REQUEST

REQUEST URI	http://www.4glte.rw/voucher/api/external_access.jsp		
PROTOCOL TYPE	HTTP	METHOD	POST
PORT	80	CONTENTS TYPE	APPLICATION/XML
I/F CYCLE	EVERY REQUEST TIME		

- REQUEST ITEM / SCHEMA

  SEND PARAMETERS (**NAME, VALUE**) : (REQ_XML, XML COMPRISING THE FOLLOWING ELEMENTS)

ELEMENT	M/O	FORMAT	DESCRIPTION
REQUEST	**M**		**REQUEST ROOT**
HEADER	**M**		**REQUEST HEADER**
SVCCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
BODY	**M**		**REQUEST PARAMETER BODY**
MSISDN	M	STRING(10)	MSISDN (CUSTMER PHONE NUMBER)
CATEGORY_CODE	M	STRING(8)	CATEGORY CODE

RESPONSE ITEM / SCHEMA

ELEMENT	M/O	FORMAT	DESCRIPTION
RESPONSE	**M**		**RESPONSE ROOT**
HEADER	**M**		**RESPONSE HEADER**
SVCCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
MSGCODE	O	STRING(8)	MESSAGE CODE
MESSAGE	O	STRING(32)	MESSAGE DESCRIPTION
BODY	**M**		**RESPONSE PARAMETER BODY**
PROD_LIST	M	STRING(2048)	THE PRODUCT LIST ORDER IS MADE UP OF THE PRODUCT CODE, THE PRODUCT NAME, AND THE PRODUCT PRICE, AND THE DELIMITER OF THE PRODUCT IS " | "
CATEGORY_CODE	M	STRING(8)	CATEGORY CODE
SO_ID	M	STRING(3)	RETAILER ID
APP_USE_YN	M	STRING(1)	APPLICATION USE OR NOT(Y/N)

MESSAGE

MESSAGE	MESSAGE DESCRIPTION

CODE	
-5002	MSISDN'S BUSINESS NO. AND TRANSFER XML'S BUSINESS NO. DO NOT MATCH.
-8001	THERE IS NO MATCHING INFORMATION FOR MSISDN OR DISTINGUISH_ID.
-8002	VIEWED CUSTOMERS ARE NOT ELIGIBLE FOR PREPAYMENT.

EXAMPLE :

YOU SHOULD SEND THE PARAMETER THAT IS SET TO THE STRING IS CONTAINING THE FOLLOWING REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML**

```
<?XML VERSION='1.0' ENCODING='UTF-8' STANDALONE='YES' ?>
<REQUEST>
<HEADER>
  <SVCODE>HP00005</SVCODE>
  <OPCODE>HPOP104</OPCODE>
  <CATEGORY_CODE>D020101</CATEGORY_CODE>
  <CHCODE>A</CHCODE>
</HEADER>
<BODY>
      <MSISDN>0771000540</MSISDN>
      <CATEGORY_CODE>D020101</CATEGORY_CODE>
</BODY>
</REQUEST>
```

- **RESPONSE XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<RESPONSE>
  <HEADER>
    <SVCODE>HP00005</SVCODE>
    <OPCODE>HPOP104</OPCODE>
    <CHCODE>A</CHCODE>
    <MSGCODE>1001</MSGCODE>
    <MESSAGE>IT HAS BEEN SUCCESSFULLY PROCESSED. PLEASE PRESS THE CHECK BALANCE BUTTO
N TO CONFIRM THE RESULTS.
    </MESSAGE>
  </HEADER>
  <BODY>
```

```
        <PROD_LIST>PD00045904,MONTHLY BASIC UNLIMITED(PHONE ONLY),0</PROD_LIST>
        <CATEGORY_CODE>D020101</CATEGORY_CODE>
        <SO_ID>00</SO_ID>
        <RESULT>OK</RESULT>
        <APP_USE_YN>Y</APP_USE_YN>
      </BODY>
</RESPONSE>
```

2.2. Auth Service

2.2.1. Request auth login (SVCCODE: HP00005, OPCODE: HPOP201)

■ SUMMARY

YOU CAN PROCEED WITH LOGIN.

■ TELECOMMUNICATION PROTOCOL FOR REQUEST

REQUEST URI	http://www.4glte.rw/voucher/api/external_access.jsp		
PROTOCOL TYPE	HTTP	METHOD	POST
PORT	80	CONTENTS TYPE	APPLICATION/XML
I/F CYCLE	EVERY REQUEST TIME		

■ REQUEST ITEM / SCHEMA

SEND PARAMETERS (**NAME, VALUE**) : (REQ_XML, XML COMPRISING THE FOLLOWING ELEMENTS)

ELEMENT	M/O	FORMAT	DESCRIPTION
REQUEST	**M**		**REQUEST ROOT**
HEADER	**M**		**REQUEST HEADER**
SVCCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
BODY	**M**		**REQUEST PARAMETER BODY**
LOGIN_ID	O	STRING(16)	RETAILER ID
LOGIN_PW	O	STRING(65)	RETAILER PASSWORD (ENCRYPTED WITH AES, 256BIT)
MSISDN	O	STRING(16)	USER MSISDN
DISTINGUISH_NO	O	STRING(65)	USER CLASSIFICATION NUMBER (NID OR TIN NUMBER OR

					DCORPORATION ID) (ENCRYPTED WITH AES, 256BIT)
		CUST_TP	O	STRING(2)	customer type A: Individual B: Self-Employee C: Foreigner D: Corporate E: Public Sector
		USER_TYPE	M	STRING(2)	DISTINGUISHED REQUEST USER TYPE 01 : INDIVIDUAL USERS 02 : RETAILER USER
		REQ_SYS_ID	M	STRING(2)	DISTINGUISHED REQUEST TYPE 01 : PIVOT ACCESS 02 : APP ACCESS 03 : USSD ACCESS

RESPONSE ITEM / SCHEMA

ELEMENT			M/O	FORMAT	DESCRIPTION
RESPONSE			**M**		**RESPONSE ROOT**
	HEADER		**M**		**RESPONSE HEADER**
		SVCCODE	M	STRING(8)	SERVICE CODE
		OPCODE	M	STRING(8)	OPERATION CODE
		CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
		MSGCODE	O	STRING(8)	MESSAGE CODE
		MESSAGE	O	STRING(32)	MESSAGE DESCRIPTION
	BODY		**M**		**REQUEST PARAMETER BODY**
		USER_TYPE	M	STRING(2)	DISTINGUISHED REQUEST USER TYPE 01 : INDIVIDUAL USERS 02 : RETAILER USER
		APP_USE_YN	M	STRING(1)	IF THIS STAFF CAN BE USED IN AP P

		SO_ID	O	STRING(4)	RETAILER CODE
		SO_NM	O	STRING(32)	RETAILER NAME
		ORG_ID	M	STRING(32)	ORGANIZATION ID
		LONGITUDE	M	STRING(12)	LONGITUDE
		LATITUDE	M	STRING(12)	LATITUDE
		SEC_PW	M	STRING(32)	PASSWORD

MESSAGE

MESSAGE CODE	MESSAGE DESCRIPTION
-5001	DO NOT MATCH LOGIN INFO.
-5004	THE STATUS PROCESS IS WAITING. PLEASE WAIT A MOMENT UNTIL SYSTEM IS NORMALIZED.
-5005	THE PROCESS WAS FAILED. PLEASE CONTACT SYSTEM ADMINISTATOR.

EXAMPLE :

YOU SHOULD SEND THE PARAMETER THAT IS SET TO THE STRING IS CONTAINING THE FOLLOWING REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML (AGENT APP)**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<REQUEST>
 <HEADER>
   <SVCODE>HP00005</SVCODE>
   <OPCODE>HPOP201</OPCODE>
   <CHCODE>A</CHCODE>
 </HEADER>
 <BODY>
      <LOGIN_ID>AIRTEL0207</LOGIN_ID>
      <LOGIN_PW>JWTERSBGCAZ3TKH2MAOZCND6UUC=</LOGIN_PW>
      <USER_TYPE>01</USER_TYPE>
      <REQ_SYS_ID>02</REQ_SYS_ID>
</BODY>
</REQUEST>
```

- **RESPONSE XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8" STANDALONE="NO"?>
<RESPONSE>
    <HEADER>
        <MESSAGE>IT HAS BEEN SUCCESSFULLY PROCESSED.</MESSAGE>
        <MSGCODE>1001</MSGCODE>
        <SVCCODE>HP00005</SVCCODE>
        <CHCODE>A</CHCODE>
        <OPCODE>HPOP201</OPCODE>
    </HEADER>
    <BODY>
        <DISTANCE>3000</DISTANCE>
        <ORG_ID>3463</ORG_ID>
        <SO_NM>BSC</SO_NM>
        <USER_TYPE>02</USER_TYPE>
        <LONGITUDE>30.046802</LONGITUDE>
        <SO_ID>10</SO_ID>
        <SEC_PW>ABC123</SEC_PW>
        <LATITUDE>-1.939587</LATITUDE>
        <APP_USE_YN>Y</APP_USE_YN>
    </BODY>
</RESPONSE>
```

2.3. Activation Service

2.3.1. Request searching price plan (SVCCODE: HP00005, OPCODE: HPOP301)

■ SUMMARY

FOR SEARCHING PRICE PLAN CAN BE AVAILABLE SIMPLE ACTIVATION

■ TELECOMMUNICATION PROTOCOL FOR REQUEST

REQUEST URI	http://www.4glte.rw/voucher/api/external_access.jsp		
PROTOCOL TYPE	HTTP	METHOD	POST
PORT	80	CONTENTS TYPE	APPLICATION/XML
I/F CYCLE	EVERY REQUEST TIME		

■ REQUEST ITEM / SCHEMA

SEND PARAMETERS (**NAME, VALUE**) : (REQ_XML, XML COMPRISING THE FOLLOWING ELEMENTS)

ELEMENT	M/O	FORMAT	DESCRIPTION

REQUEST		**M**		**REQUEST ROOT**
	HEADER	**M**		**REQUEST HEADER**
	SVCODE	M	STRING(8)	SERVICE CODE
	OPCODE	M	STRING(8)	OPERATION CODE
	CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
	BODY	**M**		**REQUEST PARAMETER BODY**
	MSISDN	M	STRING(10)	MSISDN
	ICCID	M	STRING(19)	ICCID

RESPONSE ITEM / SCHEMA

ELEMENT		M/O	FORMAT	DESCRIPTION
RESPONSE		**M**		**RESPONSE ROOT**
	HEADER	**M**		**RESPONSE HEADER**
	SVCODE	M	STRING(8)	SERVICE CODE
	OPCODE	M	STRING(8)	OPERATION CODE
	CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
	MSGCODE	O	STRING(8)	MESSAGE CODE
	MESSAGE	O	STRING(32)	MESSAGE DESCRIPTION
	BODY	**M**		**RESPONSE PARAMETER BODY**
	SO_ID	M	STRING(2)	BUSINESS NO
	PROD_CNT	M	STRING(8)	PRICE PLAN COUNT
	PROD_CD	O	STRING(10)	PRICE PLAN CODE
	PROD_NM	O	STRING(64)	PRICE PLAN NAME
	PROD_GRP	O	STRING(64)	PRODUCT TYPE (a : PREPAID, b : POSTPAID, d : PREPAIDHYBRID)

EXAMPLE :

YOU SHOULD SEND THE PARAMETER THAT IS SET TO THE STRING IS CONTAINING THE FOLLOWING REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<REQUEST>
    <HEADER>
```

```
            <SVCODE>HP00005</SVCODE>
            <OPCODE>HPOP301</OPCODE>
            <CHCODE>A</CHCODE>
        </HEADER>
        <BODY>
            <MSISDN>07XXXXXXXX</MSISDN>
            <ICCID>89XXXXXXXXXXXXXXXXX</ICCID>
             <DISTINGUISH_NO> JWTERSBGCAZ3TKH2MAOZCND6UUC=</DISTINGUISH_NO>
        </BODY>
 </REQUEST>
```

- **RESPONSE XML**

```
<RESPONSE>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
        <OPCODE>HPOP301</OPCODE>
        <CHCODE>A</CHCODE>
        <MSGCODE>XXXX</MSGCODE>
        <MESSAGE>…</MESSAGE>
    </HEADER>
    <BODY>
        <SO_ID>01</SO_ID>
        <MSISDN>07XXXXXXXX</MSISDN>
        <PROD_CNT>3</PROD_CNT>
        <PROD_NM>ORN 500MB BUNDLE(PRE),
                 ORN 20GB BUNDLE(PRE),
                 ORN 10GB BUNDLE(PRE)
        </PROD_NM>
        <PROD_CD>PD00000574,PD00000575,PD00000576</PROD_CD>
        <PROD_GRP>a,a,a</PROD_GRP>
    </BODY>
</RESPONSE>
```

2.3.2. Request for Simple activation (SVCCODE: HP00005, OPCODE: HPOP302)

■ NOTE

SIMPLE ACTIVATION IS ONLY AVAILABLE TO PREPAY CUSTOMERS.

■ SUMMARY

YOU CAN CHECK SIMPLE ACTIVATION RESULT BY USING ORDER ID.

■ TELECOMMUNICATION PROTOCOL FOR REQUEST

REQUEST URI	http://www.4glte.rw/voucher/api/external_access.jsp		
PROTOCOL TYPE	HTTP	METHOD	POST
PORT	80	CONTENTS TYPE	APPLICATION/XML
I/F CYCLE	EVERY REQUEST TIME		

■ REQUEST ITEM / SCHEMA

SEND PARAMETERS (**NAME, VALUE**) : (REQ_XML, XML COMPRISING THE FOLLOWING ELEMENTS)

ELEMENT	M/O	FORMAT	DESCRIPTION
REQUEST	**M**		**REQUEST ROOT**
HEADER	**M**		**REQUEST HEADER**
SVCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
BODY	**M**		**REQUEST PARAMETER BODY**
SO_ID	M	STRING(3)	**RETAILER CODE**
MSISDN	M	STRING(10)	MSISDN
PROD_CD	M	STRING(10)	PRICE PLAN CODE WHICH IS SELECTED FROM PRICE PLAN LIST.
LOGIN_ID	M	STRING(15)	LOGIN ID
ORG_ID	M	STRING(10)	ORGANIZATION ID
ICCID	M	STRING(64)	ICCID
CUST_ID	M	STRING(10)	CUSTOMER ID
REQ_SYS_ID	M	STRING(2)	DISTINGUISHED REQUEST TYPE 01 : PIVOT ACCESS 02 : APP ACCESS 03 : USSD ACCESS
HAND	O	STRING(1)	HEAD

			L : LEFT HAND R : RIGHT HAND
FINGERPRINT	O	STRING(64)	FINGERPRINT FLOW
PHOTO	O	STRING(64)	PHOTO FLOW
LONGITUDE	O	STRING(12)	LONGITUDE
LATITUDE	O	STRING(12)	LATITUDE
RNUMBER	O	STRING(14)	CUSTOMER'S FIRST ACTIVE NUMBER
OTP	O	STRING(6)	OTP CODE

RESPONSE ITEM / SCHEMA

ELEMENT	M/O	FORMAT	DESCRIPTION
RESPONSE	**M**		**RESPONSE ROOT**
HEADER	**M**		**RESPONSE HEADER**
SVCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
MSGCODE	O	STRING(8)	MESSAGE CODE
MESSAGE	O	STRING(32)	MESSAGE DESCRIPTION
BODY	**M**		**RESPONSE PARAMETER BODY**
RESULT	M	STRING(2048)	RESULT DETAIL MESSAGE
PROD_RATE	M	STRING(128)	PRODUCT RATE

EXAMPLE :

YOU SHOULD SEND THE PARAMETER THAT IS SET TO THE STRING IS CONTAINING THE FOLLOWING REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<REQUEST>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
        <OPCODE>HPOP302</OPCODE>
        <CHCODE>A</CHCODE>
    </HEADER>
    <BODY>
        <SO_ID>01</SO_ID>
```

```
                <MSISDN>070XXXXXXXX</MSISDN>
                <PROD_CD>PDXXXXXXXX</PROD_CD>
                <LOGIN_ID>NTELS</LOGIN_ID>
                <ORG_ID> PZ00000102</ORG_ID
                <ICCID>89XXXXXXXXXXXXXXXXX</ICCID>
                <CUST_ID>C000111222</CUST_ID>
                <REQ_SYS_ID>01</REQ_SYS_ID>
        </BODY>
    </REQUEST>
```

- **RESPONSE XML**

```
<RESPONSE>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
        <OPCODE>HPOP302</OPCODE>
        <CHCODE>A</CHCODE>
        <MSGCODE>XXXX</MSGCODE>
        <MESSAGE>…</MESSAGE>
    </HEADER>
    <BODY>
        <RESULT>SUCCESS</RESULT>
        <PROD_RATE>23000</PROD_RATE>
</BODY>
</RESPONSE>
```

EXAMPLE :

WHEN THE CUSTOMER NEED RECEIVE THE OTP CODE YOU SHOULD COORDINATE THE API **HP00005 HPOP311**. THE **HPOP311** IS THE OTP SEND API:

REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<REQUEST>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
        <OPCODE>HPOP302</OPCODE>
        <CHCODE>A</CHCODE>
    </HEADER>
    <BODY>
        <SO_ID>01</SO_ID>
```

```
                    <MSISDN>070XXXXXXXX</MSISDN>
                    <PROD_CD>PDXXXXXXXX</PROD_CD>
                    <LOGIN_ID>NTELS</LOGIN_ID>
                    <ORG_ID> PZ00000102</ORG_ID
                    <ICCID>89XXXXXXXXXXXXXXXXX</ICCID>
                    <CUST_ID>C000111222</CUST_ID>
                    <REQ_SYS_ID>01</REQ_SYS_ID>
                    <RNUMBER>250771038484</RNUMBER>
                    <OTP>140093</OTP>
            </BODY>
        </REQUEST>
```

- **RESPONSE XML**

```
<RESPONSE>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
        <OPCODE>HPOP302</OPCODE>
        <CHCODE>A</CHCODE>
        <MSGCODE>XXXX</MSGCODE>
        <MESSAGE>…</MESSAGE>
    </HEADER>
    <BODY>
        <RESULT>SUCCESS</RESULT>
        <PROD_RATE>23000</PROD_RATE>
</BODY>
</RESPONSE>
```

EXAMPLE :

WHEN THE STAFF HAVE THE WORKING RADIUS THE LONGITUDE AND LATITUDE PARAM IS MUST BE FILL IN:

REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<REQUEST>
        <HEADER>
                <SVCODE>HP00005</SVCODE>
                <OPCODE>HPOP302</OPCODE>
                <CHCODE>A</CHCODE>
```

```
        </HEADER>
        <BODY>
                <SO_ID>01</SO_ID>
                <MSISDN>070XXXXXXXX</MSISDN>
                <PROD_CD>PDXXXXXXXX</PROD_CD>
                <LOGIN_ID>NTELS</LOGIN_ID>
                <ORG_ID> PZ00000102</ORG_ID
                <ICCID>89XXXXXXXXXXXXXXXXX</ICCID>
                <CUST_ID>C000111222</CUST_ID>
                <REQ_SYS_ID>01</REQ_SYS_ID>
                <LONGITUDE>30.046802</LONGITUDE>
                <LATITUDE>-1.939587</LATITUDE>
        </BODY>
</REQUEST>
```

- **RESPONSE XML**

```
<RESPONSE>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
        <OPCODE>HPOP302</OPCODE>
        <CHCODE>A</CHCODE>
        <MSGCODE>XXXX</MSGCODE>
        <MESSAGE>…</MESSAGE>
    </HEADER>
    <BODY>
        <RESULT>SUCCESS</RESULT>
        <PROD_RATE>23000</PROD_RATE>
</BODY>
</RESPONSE>
```

EXAMPLE :

WHEN THE CUSTOMER NEED CHECK THE FINGERPRINT/PHOTO YOU CAN FILL IN THE FINGERPRINT/PHOTO/HAND PARAM:

REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<REQUEST>
        <HEADER>
```

```
            <SVCODE>HP00005</SVCODE>
            <OPCODE>HPOP302</OPCODE>
            <CHCODE>A</CHCODE>
        </HEADER>
        <BODY>
            <SO_ID>01</SO_ID>
            <MSISDN>070XXXXXXX</MSISDN>
            <PROD_CD>PDXXXXXXX</PROD_CD>
            <LOGIN_ID>NTELS</LOGIN_ID>
            <ORG_ID>  PZ00000102</ORG_ID
            <ICCID>89XXXXXXXXXXXXXXXX</ICCID>
            <CUST_ID>C000111222</CUST_ID>
            <HAND>R</HAND>
            <FINGERPRINT>IVBORW0KGGOAAAANSUHEUGAAAAEAAAABCAYAAAAFFCSJAAAA
DULEQVR42MNK+M9QDWADHGGAWJR9AWAAAABJRU5ERKJGGG==</FINGERPRINT>
            <PHOTO>IVBORW0KGGOAAAANSUHEUGAAAAEAAAABCAYAAAAFFCSJAAAADULE
QVR42MNK+M9QDWADHGGAWJR9AWAAAABJRU5ERKJGGG==</PHOTO>
        </BODY>
</REQUEST>
```

- **RESPONSE XML**

```
<RESPONSE>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
        <OPCODE>HPOP302</OPCODE>
        <CHCODE>A</CHCODE>
        <MSGCODE>XXXX</MSGCODE>
        <MESSAGE>…</MESSAGE>
    </HEADER>
    <BODY>
        <RESULT>SUCCESS</RESULT>
        <PROD_RATE>23000</PROD_RATE>
</BODY>
</RESPONSE>
```

2.3.3. Searching available bundles (SVCCODE: HP00001, OPCODE: HPOP005)

■ Summary

YOU CAN BUNDLE LIST THAT AVAILABLE FOR PURCHASE AT THE NUMBER MATCH ON MSISDN.

■ Telecommunication Protocol for Request

REQUEST URI	http://www.4glte.rw/api_access.jsp		
PROTOCOL TYPE	HTTP	METHOD	POST
PORT	80	CONTENTS TYPE	APPLICATION/XML
I/F CYCLE	EVERY REQUEST TIME		

■ Request Item / Schema

SEND PARAMETERS (**NAME, VALUE**) : (REQ_XML, XML COMPRISING THE FOLLOWING ELEMENTS)

ELEMENT	M/O	FORMAT	DESCRIPTION
REQUEST	**M**		**REQUEST ROOT**
HEADER	**M**		**REQUEST HEADER**
SVCCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
BODY	**M**		**REQUEST PARAMETER BODY**
MSISDN	M	STRING(10)	MSISDN (CUSTMER PHONE NUMBER)
ISP_CD	M	String(3)	Business No (defined by ORN)

■ Response Item / Schema

ELEMENT	M/O	FORMAT	DESCRIPTION
RESPONSE	**M**		**RESPONSE ROOT**
HEADER	**M**		**RESPONSE HEADER**
SVCCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
MSGCODE	O	STRING(8)	MESSAGE CODE
MESSAGE	O	STRING(32)	MESSAGE DESCRIPTION
BODY	**M**		**RESPONSE PARAMETER BODY**
PROD_LIST	M	STRING(2048)	THE PRODUCT LIST ORDER IS MADE UP OF THE PRODUCT CODE, THE PRODUCT NAME, AND THE PRODUCT PRICE, AND THE DELIMITER OF THE PRODUCT IS " | "

■ Example :

YOU SHOULD SEND THE PARAMETER THAT IS SET TO THE STRING IS CONTAINING THE FOLLOWING

REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- REQUEST XML

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>

<REQUEST>

 <HEADER>

   <SVCODE>HP00001</SVCODE>

   <OPCODE>HP0P005</OPCODE>
```

```
  <CHCODE>A</CHCODE>
</HEADER>
<BODY>
```

```
        <MSISDN>0771000050</MSISDN>
        <ISP_CD>62</ISP_CD>
    </BODY>
    </REQUEST>
```

- RESPONSE XML

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<RESPONSE>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
        <OPCODE>HPOP102</OPCODE>
        <CHCODE>A</CHCODE>
        <MSGCODE>XXXX</MSGCODE>
        <MESSAGE>....</MESSAGE>
    </HEADER>
    <BODY>
        <PROD_LIST>
            PD00000574,ORN    500MB    BUNDLE(PRE),2300      |
            PD00000575,ORN 10GB BUNDLE(PRE),39600
        </PROD_LIST>
    </BODY>
</RESPONSE>
```

2.3.4. Request for NIDA getCitizen API (SVCCODE: HP00005, OPCODE: HPOP303)

- SUMMARY

  GET CITIZEN INFORMATION FROM NIDA

- TELECOMMUNICATION PROTOCOL FOR REQUEST

REQUEST URI	http://www.4glte.rw/voucher/api/external_access.jsp		
PROTOCOL TYPE	HTTP	METHOD	POST
PORT	80	CONTENTS TYPE	APPLICATION/XML
I/F CYCLE	EVERY REQUEST TIME		

■ REQUEST ITEM / SCHEMA

SEND PARAMETERS (**NAME, VALUE**) : (REQ_XML, XML COMPRISING THE FOLLOWING ELEMENTS)

ELEMENT	M/O	FORMAT	DESCRIPTION
REQUEST	**M**		**REQUEST ROOT**
HEADER	**M**		**REQUEST HEADER**
SVCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
BODY	**M**		**REQUEST PARAMETER BODY**
NID	M	STRING(20)	**NATIONAL ID**
PHOTO	M	STRING(2048)	CUSTOMER PHOTO (BASE 64)

☠RESPONSE ITEM / SCHEMA

ELEMENT	M/O	FORMAT	DESCRIPTION
RESPONSE	**M**		**RESPONSE ROOT**
HEADER	**M**		**RESPONSE HEADER**
SVCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
MSGCODE	O	STRING(8)	MESSAGE CODE
MESSAGE	O	STRING(32)	MESSAGE DESCRIPTION
BODY	**M**		**RESPONSE PARAMETER BODY**
RESULT	M	STRING(1)	SUCCESS Y/N
MSG	M	STRING(128)	MESSAGE
PHOTO		STRING(2048)	PHOTO (BASE64)
SIGNATURE		STRING(2048)	CITIZEN SIGNATURE (BASE64)
NID		STRING(20)	NATIONAL ID
NAME		STRING(50)	NAME

	BIRTHDAY		STRING(20)	BIRTHDAY
	GENDER		STRING(1)	GENDER

■ EXAMPLE :

YOU SHOULD SEND THE PARAMETER THAT IS SET TO THE STRING IS CONTAINING THE FOLLOWING REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<REQUEST>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
        <OPCODE>HPOP303</OPCODE>
        <CHCODE>A</CHCODE>
    </HEADER>
    <BODY>
        <NID>1194070023836062</NID>
        <PHOTO>/9J/4AAQSKZJRGABAAEAIQA
        ....
        HWF/WCVI+/+PVHTZ9WCUZ//2Q==
        </PHOTO>
    </BODY>
</REQUEST>
```

- **RESPONSE XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<RESPONSE>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
        <OPCODE>HPOP303</OPCODE>
        <CHCODE>A</CHCODE>
        <MSGCODE>1001</MSGCODE>
        <MESSAGE>IT HAS BEEN SUCCESSFULLY PROCESSED. PLEASE PRESS THE CHECK BALANCE BUTTON TO CONFIRM THE RESULTS.
        </MESSAGE>
    </HEADER>

    <BODY>
        <RESULT>Y</RESULT>
        <MSG />
        <PHOTO>
            /9J/4AAQSKZJRGABAAEAIQAUAAD
```

```
                    ...
                    KEMXSZPEEOXYNUA3DHTY047V2ZDQDX+0CJ//2Q==
                </PHOTO>
                <SIGNATURE>
                    /9J/4AAQSKZJRGABAQEAYAB
                    ...
                    OOOOAKKKKACIIIGD//Z
                </SIGNATURE>
                <NID>1194070023836062</NID>
                <NAME>VENANTIE MUKARUTEMBYA</NAME>
                <BIRTHDAY>01/01/1940</BIRTHDAY>
                <GENDER>F</GENDER>
            </BODY>
</RESPONSE>
```


2.3.5. Request for Get company information with TIN from RRA (SVCCODE: HP00005, OPCODE: HPOP305)

■ SUMMARY

GET COMPANY INFORMATION FROM RRA WITH TIN

■ TELECOMMUNICATION PROTOCOL FOR REQUEST

REQUEST URI	http://www.4glte.rw/voucher/api/external_access.jsp		
PROTOCOL TYPE	HTTP	METHOD	POST
PORT	80	CONTENTS TYPE	APPLICATION/XML
I/F CYCLE	EVERY REQUEST TIME		

■ REQUEST ITEM / SCHEMA

SEND PARAMETERS (**NAME, VALUE**) : (REQ_XML, XML COMPRISING THE FOLLOWING ELEMENTS)

ELEMENT	M/O	FORMAT	DESCRIPTION
REQUEST	**M**		**REQUEST ROOT**
HEADER	**M**		**REQUEST HEADER**
SVCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
BODY	**M**		**REQUEST PARAMETER BODY**
TIN	M	STRING(20)	TIN

	REGR_ID	M	STRING(20)	REGISTER ID

■ RESPONSE ITEM / SCHEMA

ELEMENT		M/O	FORMAT	DESCRIPTION
RESPONSE		**M**		**RESPONSE ROOT**
	HEADER	**M**		**RESPONSE HEADER**
	SVCODE	M	STRING(8)	SERVICE CODE
	OPCODE	M	STRING(8)	OPERATION CODE
	CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
	MSGCODE	O	STRING(8)	MESSAGE CODE
	MESSAGE	O	STRING(32)	MESSAGE DESCRIPTION
	BODY	**M**		**RESPONSE PARAMETER BODY**
	MESSAGECODE	M	STRING(5)	MESSAGECODE
	MESSAGEDESCRIPTION	M	STRING(128)	MESSAGEDESCRIPTION
	SUPPLIERNAME	O	STRING(128)	SUPPLIERNAME
	EMAILADDRESS	O	STRING(128)	EMAIL ADDRESS
	PHONENUMBER	O	STRING(20)	PHONE NUMBER
	PROVINCE	O	STRING(50)	PROVINCE
	DETAILEDADDRESS	O	STRING(50)	DETAILEDADDRESS
	DISTRICT	O	STRING(50)	DISTRICT
	CATEGORY	O	STRING(50)	CATEGORY
	CELL	O	STRING(50)	CELL
	SECTOR	O	STRING(50)	SECTOR

■ EXAMPLE :

YOU SHOULD SEND THE PARAMETER THAT IS SET TO THE STRING IS CONTAINING THE FOLLOWING REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<REQUEST>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
        <OPCODE>HPOP305</OPCODE>
        <CHCODE>A</CHCODE>
    </HEADER>
    <BODY>
        <TIN>999999999</TIN>
```

```
                <REGR_ID>CCBS</ REGR_ID >
        </BODY>
 </REQUEST>
```

- **RESPONSE XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<RESPONSE>
        <HEADER>
                <SVCODE>HP00005</SVCODE>
                <OPCODE>HPOP305</OPCODE>
                <CHCODE>A</CHCODE>
                <MSGCODE>1001</MSGCODE>
                <MESSAGE>IT HAS BEEN SUCCESSFULLY PROCESSED. PLEASE PRESS THE CHECK BALANCE BUTTON TO CONFIRM THE RESULTS.
                </MESSAGE>
        </HEADER>

        <BODY>
                <SUPPLIERNAME>KT RWANDA NETWORKS LTD</SUPPLIERNAME>
                <CELL>KIMIHURURA</CELL>
                <MANAGERNAMES>SANDRA RWAMUSHAIJA</MANAGERNAMES>
                <EMAILADDRESS>ROBERT.KARANGWA@KTRN.RW</EMAILADDRESS>
                <PHONENUMBER>729200100</PHONENUMBER>
                <PROVINCE>KIGALI CITY</PROVINCE>
                <DETAILEDADDRESS>5440</DETAILEDADDRESS>
                <DISTRICT>GASABO</DISTRICT>
                <MESSAGECODE>1</MESSAGECODE>
                <MESSAGEDESCRIPTION>RRA SUCCESS MESSAGE - SUPPLIER DETAILS FOUND</MESSAGEDESCRIPTION>
                <CATEGORY>LARGE TAXPAYER</CATEGORY>
                <VILLAGE>RUGARAMA</VILLAGE>
                <SECTOR>KIMIRURA</SECTOR>
                <SUPPLIERTIN>103077243</SUPPLIERTIN>
        </BODY>
</RESPONSE>
```

2.3.6. Searching Subscriber Count (SVCCODE: HP00005, OPCODE: HPOP310)

■ SUMMARY

YOU CAN CHECK THE SUBS COUNT UNDER THIS CERT.

■ TELECOMMUNICATION PROTOCOL FOR REQUEST

REQUEST URI	http://www.4glte.rw/voucher/api/external_access.jsp		
PROTOCOL TYPE	HTTP	METHOD	POST
PORT	80	CONTENTS TYPE	APPLICATION/XML
I/F CYCLE	EVERY REQUEST TIME		

■ REQUEST ITEM / SCHEMA

SEND PARAMETERS (**NAME, VALUE**) : (REQ_XML, XML COMPRISING THE FOLLOWING ELEMENTS)

ELEMENT	M/O	FORMAT	DESCRIPTION
REQUEST	**M**		**REQUEST ROOT**
HEADER	**M**		**REQUEST HEADER**
SVCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
BODY	**M**		**REQUEST PARAMETER BODY**
SO_ID	M	STRING(3)	**RETAILER CODE**
CUST_TYPE	M	STRING(1)	THE CUSTOMER TYPE: A : INDIVIDUAL CUSTOMER B : CORPORATE CUSTOMER C : FOREIGNER CUSTOMER
CUST_NM	M	STRING(64)	CUSTOMER'S NAME
CORP_REG_NO	M	STRING(128)	ENCRYPTION CERT NUMBER

■ RESPONSE ITEM / SCHEMA

ELEMENT	M/O	FORMAT	DESCRIPTION
RESPONSE	**M**		**RESPONSE ROOT**
HEADER	**M**		**RESPONSE HEADER**
SVCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
MSGCODE	O	STRING(8)	MESSAGE CODE
MESSAGE	O	STRING(32)	MESSAGE DESCRIPTION
BODY	**M**		**RESPONSE PARAMETER BODY**
SUBSCOUNT	M	STRING(32)	THE SUBS COUNT OF THIS CERT
FISRTNUMBER	M	STRING(9)	THE CUSTOMER'S FIRST NUMBER

■ EXAMPLE :

WHEN YOU WANT TO KNOW THE NUMBER UNDER THIS CERT OR THE FIRST NUMBER OF THIS CUST:

REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<REQUEST>
    <HEADER>
        <SVCCODE>HP00005</SVCCODE>
        <OPCODE>HPOP310</OPCODE>
        <CHCODE>A</CHCODE>
    </HEADER>
    <BODY>
        <SO_ID>10</SO_ID>
        <CUST_TP>A</CUST_TP>
        <CUST_NM>DUSENGE AUDACE</CUST_NM>

      <CORP_REG_NO>E1U%2FH9JCTG7PCAJV0IMGWKE4T12KD%2FWAR1P9ADEWNT4%3D</CO
      RP_REG_NO>
    </BODY>
</REQUEST>
```

- **RESPONSE XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8" STANDALONE="NO"?>
<RESPONSE>
    <HEADER>
        <MESSAGE>IT HAS BEEN SUCCESSFULLY PROCESSED.</MESSAGE>
        <MSGCODE>1001</MSGCODE>
        <SVCCODE>HP00005</SVCCODE>
        <CHCODE>A</CHCODE>
        <OPCODE>HPOP310</OPCODE>
    </HEADER>
    <BODY>
        <SUBSCOUNT>1</SUBSCOUNT>
        <FISRTNUMBER>771005888</FISRTNUMBER>
    </BODY>
</RESPONSE>
```

2.3.7. Get OTP Code (SVCCODE: HP00005, OPCODE: HPOP311)

■ SUMMARY

YOU CAN SEND THE OTP FOR NUMBER.

■ TELECOMMUNICATION PROTOCOL FOR REQUEST

REQUEST URI	http://www.4glte.rw/voucher/api/external_access.jsp		
PROTOCOL TYPE	HTTP	METHOD	POST
PORT	80	CONTENTS TYPE	APPLICATION/XML
I/F CYCLE	EVERY REQUEST TIME		

■ REQUEST ITEM / SCHEMA

SEND PARAMETERS (**NAME, VALUE**) : (REQ_XML, XML COMPRISING THE FOLLOWING ELEMENTS)

ELEMENT	M/O	FORMAT	DESCRIPTION
REQUEST	**M**		**REQUEST ROOT**
HEADER	**M**		**REQUEST HEADER**
SVCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
BODY	**M**		**REQUEST PARAMETER BODY**
SO_ID	M	STRING(3)	**RETAILER CODE**
OTPTYPE	M	STRING(32)	THE EVENT TYPE: KTRN_NEW_CONNECTION_OTP
RNUMBER	M	STRING(12)	CUSTOMER'S FIRST NUMBER

■ RESPONSE ITEM / SCHEMA

ELEMENT	M/O	FORMAT	DESCRIPTION
RESPONSE	**M**		**RESPONSE ROOT**
HEADER	**M**		**RESPONSE HEADER**
SVCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
MSGCODE	O	STRING(8)	MESSAGE CODE
MESSAGE	O	STRING(32)	MESSAGE DESCRIPTION
BODY	**M**		**RESPONSE PARAMETER BODY**

■ EXAMPLE :

WHEN YOU WANT TO SEND ONE OTP FOR THE NUMBER:

REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML**

```
<REQUEST>
    <HEADER>
        <SVCCODE>HP00005</SVCCODE>
        <OPCODE>HPOP311</OPCODE>
        <CHCODE>A</CHCODE>
    </HEADER>
    <BODY>
        <SO_ID>01</SO_ID>
        <OTPTYPE>KTRN_NEW_CONNECTION_OTP</OTPTYPE>
        <RNUMBER>250771000251</RNUMBER>

    </BODY>
</REQUEST>
```

- **RESPONSE XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8" STANDALONE="NO"?>
<RESPONSE>
    <HEADER>
        <MESSAGE>IT HAS BEEN SUCCESSFULLY PROCESSED.</MESSAGE>
        <MSGCODE>1001</MSGCODE>
        <SVCCODE>HP00005</SVCCODE>
        <CHCODE>A</CHCODE>
        <OPCODE>HPOP311</OPCODE>
    </HEADER>
    <BODY/>
</RESPONSE>
```

2.3.8. Verifying Fingerprint (SVCCODE: HP00005, OPCODE: HPOP312)

■ SUMMARY

YOU CAN CHECK IF THIS CUSTOMER'S FINGERPRINT IS RIGHT.

■ TELECOMMUNICATION PROTOCOL FOR REQUEST

REQUEST URI	http://www.4glte.rw/voucher/api/external_access.jsp		
PROTOCOL TYPE	HTTP	METHOD	POST
PORT	80	CONTENTS TYPE	APPLICATION/XML
I/F CYCLE	EVERY REQUEST TIME		

■ REQUEST ITEM / SCHEMA

SEND PARAMETERS (**NAME, VALUE**) : (REQ_XML, XML COMPRISING THE FOLLOWING ELEMENTS)

ELEMENT	M/O	FORMAT	DESCRIPTION
REQUEST	**M**		**REQUEST ROOT**
HEADER	**M**		**REQUEST HEADER**
SVCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
BODY	**M**		**REQUEST PARAMETER BODY**
SO_ID	M	STRING(3)	**RETAILER CODE**
CUST_TP		STRING(1)	THE CUSTOMER TYPE: A : INDIVIDUAL CUSTOMER B : CORPORATE CUSTOMER C : FOREIGNER CUSTOMER
CUST_NM	M	STRING(64)	CUSTOMER'S NAME
CORP_REG_NO	M	STRING(128)	ENCRYPTION CERT NUMBER
HAND	M	STRING(1)	HEAD L : LEFT HAND R : RIGHT HAND
FINGERPRINT	M	STRING(12)	THE FINGERPRINT FLOW

■ RESPONSE ITEM / SCHEMA

ELEMENT	M/O	FORMAT	DESCRIPTION
RESPONSE	**M**		**RESPONSE ROOT**
HEADER	**M**		**RESPONSE HEADER**
SVCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
MSGCODE	O	STRING(8)	MESSAGE CODE

	MESSAGE	O	STRING(32)	MESSAGE DESCRIPTION
BODY		**M**		**RESPONSE PARAMETER BODY**
RESULT		**M**	STRING(1)	F : FALSE S : SUCCESS

■ EXAMPLE :

WHEN YOU WANT TO CHECK IF THE FINGERPRINT IS RIGHT:

REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML**

```
<REQUEST>
    <HEADER>
        <SVCCODE>HP00005</SVCCODE>
        <OPCODE>HPOP312</OPCODE>
        <CHCODE>A</CHCODE>
    </HEADER>
    <BODY>
        <SO_ID>10</SO_ID>
        <CUST_TP>A</CUST_TP>
        <CUST_NM>GUOXUEHU</CUST_NM>

      <CORP_REG_NO>E1U%2FH9JCTG7PCAJV0IMGWKE4T12KD%2FWAR1P9ADEWNT4%3D</CO
      RP_REG_NO>
        <HAND>R</HAND>

      <FINGERPRINT>IVBORW0KGGOAAAANSUHEUGAAAAEAAAABCAYAAAAFFCSJAAAADULEQV
      R42MNK+M9QDWADHGGAWJR9AWAAAABJRU5ERKJGGG==</FINGERPRINT>
    </BODY>
</REQUEST>
```

- **RESPONSE XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8" STANDALONE="NO"?>
<RESPONSE>
    <HEADER>
        <MESSAGE>IT HAS BEEN SUCCESSFULLY PROCESSED.</MESSAGE>
        <MSGCODE>1001</MSGCODE>
        <SVCCODE>HP00005</SVCCODE>
        <CHCODE>A</CHCODE>
        <OPCODE>HPOP312</OPCODE>
```

```
    </HEADER>
    <BODY>
        <RESULT>S</RESULT>
    </BODY>
</RESPONSE>
```

2.3.9. Encrypt/Decrypt Sensitive PWD/CERT (SVCCODE: HP00005, OPCODE: HPOP313)

■ SUMMARY

YOU CAN ENCRYPT/DECRYPT THE PASSWORD/CERT NUMBER.

■ TELECOMMUNICATION PROTOCOL FOR REQUEST

REQUEST URI	http://www.4glte.rw/voucher/api/external_access.jsp		
PROTOCOL TYPE	HTTP	METHOD	POST
PORT	80	CONTENTS TYPE	APPLICATION/XML
I/F CYCLE	EVERY REQUEST TIME		

■ REQUEST ITEM / SCHEMA

SEND PARAMETERS (**NAME, VALUE**) : (REQ_XML, XML COMPRISING THE FOLLOWING ELEMENTS)

ELEMENT	M/O	FORMAT	DESCRIPTION
REQUEST	**M**		**REQUEST ROOT**
HEADER	**M**		**REQUEST HEADER**
SVCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
BODY	**M**		**REQUEST PARAMETER BODY**
PASSWORD	M	STRING(256)	THE EN/DECRYPT STRING
OPERATION_TYPE	M	STRING(2)	IF YOU ENCODE NO NEED TO BE FILLED IF YOU DECODE YOU SHOULD FIL L IN “DP” DP : DECODE

■ RESPONSE ITEM / SCHEMA

ELEMENT	M/O	FORMAT	DESCRIPTION
RESPONSE	**M**		**RESPONSE ROOT**

HEADER		**M**		**RESPONSE HEADER**
	SVCODE	M	STRING(8)	SERVICE CODE
	OPCODE	M	STRING(8)	OPERATION CODE
	CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
	MSGCODE	O	STRING(8)	MESSAGE CODE
	MESSAGE	O	STRING(32)	MESSAGE DESCRIPTION
BODY		**M**		**RESPONSE PARAMETER BODY**
RESULT		**M**	STRING(2 5 6)	EN/DECRYPT RESULT

■ EXAMPLE :

WHEN YOU WANT TO ENCRYPT PASSWORD/CERT NUMBER:

REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<REQUEST>
    <HEADER>
        <SVCCODE>HP00005</SVCCODE>
        <OPCODE>HPOP313</OPCODE>
        <CHCODE>A</CHCODE>
    </HEADER>
    <BODY>
        <PASSWORD>1298180005772048</PASSWORD>
        <OPERATION_TYPE></OPERATION_TYPE>
    </BODY>
</REQUEST>
```

- **RESPONSE XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8" STANDALONE="NO"?>
<RESPONSE>
    <HEADER>
        <MESSAGE>IT HAS BEEN SUCCESSFULLY PROCESSED.</MESSAGE>
        <MSGCODE>1001</MSGCODE>
        <SVCCODE>HP00005</SVCCODE>
        <CHCODE>A</CHCODE>
        <OPCODE>HPOP313</OPCODE>
    </HEADER>
```

```
    <BODY>
        <RESULT>U9ZRHIZRERVMX6XOYFXV%2FDKSCO1ZYLV5ZZX%2F8YPQAJY%3D</RESULT>
    </BODY>
</RESPONSE>
```

■ EXAMPLE :

WHEN YOU WANT TO DECRYPT PASSWORD/CERT NUMBER:

REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<REQUEST>
    <HEADER>
        <SVCCODE>HP00005</SVCCODE>
        <OPCODE>HPOP313</OPCODE>
        <CHCODE>A</CHCODE>
    </HEADER>
    <BODY>
        <!-- ENCODE -->
        <!-- <PASSWORD>1298180005772048</PASSWORD>
        <OPERATION_TYPE></OPERATION_TYPE> -->
        <!-- DECODE -->

<PASSWORD>E1U%2FH9JCTG7PCAJV0IMGWKE4T12KD%2FWAR1P9ADEWNT4%3D</PASSWORD
>
        <OPERATION_TYPE>DP</OPERATION_TYPE>

    </BODY>
</REQUEST>
```

- **RESPONSE XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8" STANDALONE="NO"?>
<RESPONSE>
    <HEADER>
        <MESSAGE>IT HAS BEEN SUCCESSFULLY PROCESSED.</MESSAGE>
        <MSGCODE>1001</MSGCODE>
        <SVCCODE>HP00005</SVCCODE>
        <CHCODE>A</CHCODE>
        <OPCODE>HPOP313</OPCODE>
```

```
        </HEADER>
        <BODY>
            <RESULT>1198180005772048</RESULT>
        </BODY>
</RESPONSE>
```

2.4. Purchase Product Service

2.4.1. Request Bundle Purchase (SVCCODE: HP00005, OPCODE: HPOP401)

■ SUMMARY

YOU CAN PURCHASE THE BUNDLE PRODUCT.

■ TELECOMMUNICATION PROTOCOL FOR REQUEST

REQUEST URI	http://www.4glte.rw/voucher/api/external_access.jsp		
PROTOCOL TYPE	HTTP	METHOD	POST
PORT	80	CONTENTS TYPE	APPLICATION/XML
I/F CYCLE	EVERY REQUEST TIME		

■ REQUEST ITEM / SCHEMA

SEND PARAMETERS (**NAME, VALUE**) : (REQ_XML, XML COMPRISING THE FOLLOWING ELEMENTS)

ELEMENT	M/O	FORMAT	DESCRIPTION
REQUEST	M		**REQUEST ROOT**
HEADER	M		**REQUEST HEADER**
SVCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
BODY	M		**REQUEST PARAMETER BODY**
SO_ID	M	STRING(3)	RETAILER CODE
PROD_CD	M	STRING(10)	SELECTED PRODUCT CODE
MSISDN	M	STRING(10)	SERVICE TEL NUMBER
PAYMENT_TYPE	M	STRING(4)	PAYMENT_TYPE (01 : CASH 02 : MOBILE MONEY

				03 : CREDIT CARD 04 : BANK TRANSFER)
	CREDIT_TYPE	O	STRING(64)	CREDIT TYPE (01 : ENCISIA 02 : 3GDP 03 : PIVOT)
	REFID	O	STRING(64)	REFERENCE ID (ESICIA)
	TID	O	STRING(64)	TID (ESICIA)
	PAYMENT_AMOUNT	O	STRING(64)	PAID AMOUNT (ENSICIA)
	AUTHKEY	O	STRING(128)	APPROVAL NUMBER (ENSICIA)
	REQ_SYS_ID	M	STRING(2)	DISTINGUISHED REQUEST TYPE 01 : PIVOT ACCESS 02 : APP ACCESS 03 : USSD ACCESS

■ RESPONSE ITEM / SCHEMA'

ELEMENT			M/O	FORMAT	DESCRIPTION
RESPONSE			**M**		**RESPONSE ROOT**
	HEADER		**M**		**RESPONSE HEADER**
		SVCODE	M	STRING(7)	SERVICE CODE
		OPCODE	M	STRING(7)	OPERATION CODE
		CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
		MSGCODE	O	STRING(8)	MESSAGE CODE
		MESSAGE	O	STRING(32)	MESSAGE DESCRIPTION
	BODY		**M**		**RESPONSE PARAMETER BODY**
		MSISDN	M	STRING(10)	SERVICE TEL NUMBER
		PROD_NM	M	STRING(50)	PRODUCT NAME
		PROD_GRP	M	STRING(1)	PRODUCT GROUP (a : prepaid, b : postpaid d : prepaidhybrid)
		PAYMENT_TYPE	M	STRING(4)	PAYMENT_TYPE (01 : CASH 02 : MOBILE MONEY 03 : CREDIT CARD 04 : BANK TRANSFER)
		CREDIT_TYPE	O	STRING(64)	CREDIT TYPE (01 : ENCISIA 02 : 3GDP

					03 : PIVOT)

■ EXAMPLE :

YOU SHOULD SEND THE PARAMETER THAT IS SET TO THE STRING IS CONTAINING THE FOLLOWING REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<REQUEST>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
        <OPCODE>HPOP401</OPCODE>
        <CHCODE>A</CHCODE>
    </HEADER>
    <BODY>
        <SO_ID>02</SO_ID>
        <PROD_CD>PD00000574</PROD_CD>
        <MSISDN>0771000050</MSISDN>
        <PAYMENT_TYPE>01</PAYMENT_TYPE>
        <REQ_SYS_ID>01</REQ_SYS_ID>
    </BODY>
</REQUEST>
```

- **RESPONSE XML**

```
<RESPONSE>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
        <OPCODE>HPOP401</OPCODE>
        <CHCODE>A</CHCODE>
        <MSGCODE>XXXX</MSGCODE>
        <MESSAGE>…</MESSAGE>
    </HEADER>
    <BODY>
        <MSISDN>SUCCESS</MSISDN>
        <PROD_NM>ISP MUP 3GB</PROD_NM>
        <PROD_GRP>a</PROD_GRP>
        <PAYMENT_TYPE>01</PAYMENT_TYPE>
    </BODY>
```

</RESPONSE>

2.4.2. Request E-Wallet Amount (SVCCODE: HP00005, OPCODE: HPOP402)

■ SUMMARY

YOU CAN PURCHASE THE BUNDLE PRODUCT.

■ TELECOMMUNICATION PROTOCOL FOR REQUEST

REQUEST URI	http://www.4glte.rw/voucher/api/external_access.jsp		
PROTOCOL TYPE	HTTP	METHOD	POST
PORT	80	CONTENTS TYPE	APPLICATION/XML
I/F CYCLE	EVERY REQUEST TIME		

■ REQUEST ITEM / SCHEMA

SEND PARAMETERS (**NAME, VALUE**) : (REQ_XML, XML COMPRISING THE FOLLOWING ELEMENTS)

ELEMENT	M/O	FORMAT	DESCRIPTION
REQUEST	M		**REQUEST ROOT**
HEADER	M		**REQUEST HEADER**
SVCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
BODY	M		**REQUEST PARAMETER BODY**
SO_ID	M	STRING(3)	RETAILER CODE
PROD_CD	M	STRING(10)	SELECTED PRODUCT CODE

■ RESPONSE ITEM / SCHEMA'

ELEMENT	M/O	FORMAT	DESCRIPTION
RESPONSE	**M**		**RESPONSE ROOT**
HEADER	**M**		**RESPONSE HEADER**
SVCODE	M	STRING(7)	SERVICE CODE
OPCODE	M	STRING(7)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
MSGCODE	O	STRING(8)	MESSAGE CODE
MESSAGE	O	STRING(32)	MESSAGE DESCRIPTION

BODY		**M**		**RESPONSE PARAMETER BODY**
	RESULT	M	STRING(1)	Y : CAN PURCAHSE N : CANNOT PURCHASE

■ EXAMPLE :

YOU SHOULD SEND THE PARAMETER THAT IS SET TO THE STRING IS CONTAINING THE FOLLOWING REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<REQUEST>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
        <OPCODE>HPOP402</OPCODE>
        <CHCODE>A</CHCODE>
    </HEADER>
    <BODY>
        <SO_ID>02</SO_ID>
        <PROD_CD>PD00000574</PROD_CD>
    </BODY>
</REQUEST>
```

- **RESPONSE XML**

```
<RESPONSE>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
        <OPCODE>HPOP402</OPCODE>
        <CHCODE>A</CHCODE>
        <MSGCODE>XXXX</MSGCODE>
        <MESSAGE>…</MESSAGE>
    </HEADER>
    <BODY>
        <RESULT>Y</RESULT>
    </BODY>
</RESPONSE>
```

2.4.3. Request Bundle Purchase Checking (SVCCODE: HP00005, OPCODE: HPOP403)

■ SUMMARY

YOU CAN CHECK THE BUNDLE PRODUCT PURCHASE OR NOT

■ TELECOMMUNICATION PROTOCOL FOR REQUEST

REQUEST URI	http://www.4glte.rw/voucher/api/external_access.jsp		
PROTOCOL TYPE	HTTP	METHOD	POST
PORT	80	CONTENTS TYPE	APPLICATION/XML
I/F CYCLE	EVERY REQUEST TIME		

■ REQUEST ITEM / SCHEMA

SEND PARAMETERS (**NAME, VALUE**) : (REQ_XML, XML COMPRISING THE FOLLOWING ELEMENTS)

ELEMENT	M/O	FORMAT	DESCRIPTION
REQUEST	M		**REQUEST ROOT**
HEADER	M		**REQUEST HEADER**
SVCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
BODY	M		**REQUEST PARAMETER BODY**
SO_ID	M	STRING(3)	RETAILER CODE
PROD_CD	M	STRING(10)	SELECTED PRODUCT CODE
MSISDN	O	STRING(10)	SERVICE TEL NUMBER
ICCID	O	STRING(19)	ICCID
PAYMENT_TYPE	M	STRING(4)	PAYMENT_TYPE (01 : CASH 02 : MOBILE MONEY 03 : CREDIT CARD 04 : BANK TRANSFER)
CREDIT_TYPE	O	STRING(64)	CREDIT TYPE (01 : ENCISIA 02 : 3GDP 03 : PIVOT)

REFID	O	STRING(64)	REFERENCE ID (ESICIA)
TID	O	STRING(64)	TID (ESICIA)
PAYMENT_AMOUNT	O	STRING(64)	PAID AMOUNT (ENSICIA)
AUTHKEY	O	STRING(128)	APPROVAL NUMBER (ENSICIA)
REQ_SYS_ID	M	STRING(2)	DISTINGUISHED REQUEST TYPE 01 : PIVOT ACCESS 02 : APP ACCESS 03 : USSD ACCESS

■ RESPONSE ITEM / SCHEMA'

ELEMENT	M/O	FORMAT	DESCRIPTION
RESPONSE	**M**		**RESPONSE ROOT**
HEADER	**M**		**RESPONSE HEADER**
SVCODE	M	STRING(7)	SERVICE CODE
OPCODE	M	STRING(7)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
MSGCODE	O	STRING(8)	MESSAGE CODE
MESSAGE	O	STRING(32)	MESSAGE DESCRIPTION
BODY	**M**		**RESPONSE PARAMETER BODY**
RESULT	M	STRING(1)	Y

■ EXAMPLE :

YOU SHOULD SEND THE PARAMETER THAT IS SET TO THE STRING IS CONTAINING THE FOLLOWING REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<REQUEST>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
        <OPCODE>HPOP403</OPCODE>
        <CHCODE>A</CHCODE>
    </HEADER>
    <BODY>
        <SO_ID>02</SO_ID>
        <PROD_CD>PD00000574</PROD_CD>
        <MSISDN>0771000050</MSISDN>
        <PAYMENT_TYPE>01</PAYMENT_TYPE>
```

```
            <REQ_SYS_ID>01</REQ_SYS_ID>
        </BODY>
    </REQUEST>
```

- **RESPONSE XML**

```
<RESPONSE>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
        <OPCODE>HPOP403</OPCODE>
        <CHCODE>A</CHCODE>
        <MSGCODE>1001</MSGCODE>
        <MESSAGE>IT HAS BEEN SUCCESSFULLY PROCESSED. PLEASE PRESS THE CHECK BALANCE
BUTTON TO CONFIRM THE RESULTS.</MESSAGE>
    </HEADER>
    <BODY>
        <RESULT>Y</RESULT>
    </BODY>
</RESPONSE>

<RESPONSE>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
        <OPCODE>HPOP403</OPCODE>
        <CHCODE>A</CHCODE>
        <MSGCODE>-7008</MSGCODE>
        <MESSAGE>YOU CANNOT PURCHASE IT BECAUSE YOU ALREADY BOUGHT TWO BUNDLE
PRODUCTS.</MESSAGE>
    </HEADER>
</RESPONSE>
```

2.4.4. Request BundlePurchase (SVCCODE: HP00002, OPCODE: HPOP004)

■ Summary

YOU CAN PURCHASE THE BUNDLE PRODUCT.

■ Telecommunication PROTOCOL FOR REQUEST

REQUEST URI	http://www.4glte.rw/api_access.jsp

PROTOCOL TYPE	HTTP	METHOD	POST
PORT	80	CONTENTS TYPE	APPLICATION/XML
I/F CYCLE	EVERY REQUEST TIME		

■ Request ITEM / SCHEMA

SEND PARAMETERS (**NAME, VALUE**) : (REQ_XML, XML COMPRISING THE FOLLOWING ELEMENTS)

ELEMENT	M/O	FORMAT	DESCRIPTION
REQUEST	M		**REQUEST ROOT**
HEADER	M		**REQUEST HEADER**
SVCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
BODY	M		**REQUEST PARAMETER BODY**
MSISDN	M	String(15)	MSISDN
ISP_CD	M	String(3)	Business No (defined by ORN)
RCHRG_TP	M	String	Recharge Type 500 : Purchase Bundle
RCHRG_STAT	M	String	Recharge Status 10 : Purchase

		PROD_CD	M	STRING(10)	SELECTED PRODUCT CODE

■ Response ITEM / SCHEMA'

ELEMENT			M/O	FORMAT	DESCRIPTION
RESPONSE			**M**		**RESPONSE ROOT**
	HEADER		**M**		**RESPONSE HEADER**
		SVCODE	M	STRING(7)	SERVICE CODE
		OPCODE	M	STRING(7)	OPERATION CODE
		CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
		MSGCODE	O	STRING(8)	MESSAGE CODE
		MESSAGE	O	STRING(32)	MESSAGE DESCRIPTION
	BODY		**M**		**RESPONSE PARAMETER BODY**
		RESULT	M	STRING(2048)	OK : SUCCESS

■ Example :

YOU SHOULD SEND THE PARAMETER THAT IS SET TO THE STRING IS CONTAINING THE FOLLOWING REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- REQUEST XML

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>

<REQUEST>

    <HEADER>

        <SVCODE>HP00002</SVCODE>

        <OPCODE>HPOP004</OPCODE>

        <CHCODE>A</CHCODE>

    </HEADER>

    <BODY>

        <MSISDN>0771000050</MSISDN>

        <SO_ID>62</SO_ID>

        <RCHRG_TP>500</RCHRG_TP>
        <RCHRG_STAT>10</RCHRG_STAT>
        <PROD_CD>PD00000574</PROD_CD>
```

```
        </BODY>
    </REQUEST>
```

2.5. Subscribe Customer Info

2.5.1. Subscribe Customer Info (SVCCODE:HP00005 , OPCODE:HPOP501)

■ SUMMARY

FOR NEW CUSTOMER REGISTRATION

■ TELECOMMUNICATION PROTOCOL FOR REQUEST

REQUEST URI	http://www.4glte.rw/voucher/api/external_access.jsp		
PROTOCOL TYPE	HTTP	METHOD	POST
PORT	80	CONTENTS TYPE	APPLICATION/XML
I/F CYCLE	EVERY REQUEST TIME		

■ REQUEST ITEM / SCHEMA

SEND PARAMETERS (**NAME, VALUE**) : (REQ_XML, XML COMPRISING THE FOLLOWING ELEMENTS)

ELEMENT			M/O	FORMAT	DESCRIPTION
REQUEST			**M**		**REQUEST ROOT**
	HEADER		**M**		**REQUEST HEADER**
		SVCODE	M	STRING(8)	SERVICE CODE
		OPCODE	M	STRING(8)	OPERATION CODE
		CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
	BODY		**M**		**REQUEST PARAMETER BODY**
		SO_ID	O	STRING(3)	RETAILER ID
		LOGIN_ID	O	STRING(15)	LOGIN ID
		ORG_ID	M	STRING(10)	ORGANIZATION ID
		CUST_TP	M	STRING(2)	customer type A: Individual B: Self-Employee C: Foreigner D: Corporate E: Public Sector
		CUST_NM	M	STRING(50)	Customer name

		CORP_REG_NO	M	STRING(50)	Corporation number (CUST_TP: A,B,C,D Mandatory) (ENCRYPTED WITH AES, 256BIT)
		BIZ_REG_NO	M	STRING(10)	Business number (CUST_TP: B,D,E mandatory)
		BIRTH_DAY	M	STRING(8)	Customer birth day (A,B,C mandatory) EX) 20180101
		EMAIL	O	STRING(30)	EMAIL
		PHONE_NUM	O	STRING(20)	PHONE NUMBER
		ADDR_DTL	M	STRING(300)	ADDRESS DETAIL
		LONGITUDE	O	STRING(12)	LONGITUDE
		LATITUDE	O	STRING(12)	LATITUDE

■ RESPONSE ITEM / SCHEMA

ELEMENT			M/O	FORMAT	DESCRIPTION
RESPONSE			**M**		**RESPONSE ROOT**
	HEADER		**M**		**RESPONSE HEADER**
		SVCODE	M	STRING(8)	SERVICE CODE
		OPCODE	M	STRING(8)	OPERATION CODE
		CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
		MSGCODE	O	STRING(8)	MESSAGE CODE
		MESSAGE	O	STRING(32)	MESSAGE DESCRIPTION
	BODY		**M**		**RESPONSE PARAMETER BODY**
	CUST_ID		M	STRING(10)	CUSTOMER ID

■ EXAMPLE :

YOU SHOULD SEND THE PARAMETER THAT IS SET TO THE STRING IS CONTAINING THE FOLLOWING

- **REQUEST XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<REQUEST>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
```

```
                <OPCODE>HPOP501</OPCODE>
                <CHCODE>A</CHCODE>
            </HEADER>
            <BODY>
                <SO_ID>02</SO_ID>
                <LOGIN_ID>TIGO0203</LOGIN_ID>
                <ORG_ID>NZ00023</ORG_ID>
                <CUST_TP>A</CUST_TP>
                <CUST_NM>DAVID</CUST_NM>
                <CORP_REG_NO>431DFDAS1212123</CORG_REG_NO>
                <BIRTH_DAY>20100101</BIRTH_DAY>
<EMAIL>KTRN@GMAIL.COM</EMAIL>
                <PHONE_NUM>0708897789</PHONE_NUM>
                <ADDR_DTL>NYARUTARAMA</ADDR_DTL>
            </BODY>
        </REQUEST>
```

- **RESPONSE XML**

```
<RESPONSE>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
        <OPCODE>HPOP501</OPCODE>
        <CHCODE>A</CHCODE>
        <MSGCODE>1001</MSGCODE>
        <MESSAGE>…</MESSAGE>
    </HEADER>
    <BODY>
    </BODY>
</RESPONSE>
```

2.5.2. Searching Customer Info (SVCCODE : HP00005, OPCODE : HPOP502)

■ SUMMARY

FOR SERCHING CUSTOMER INFOMATION

■ TELECOMMUNICATION PROTOCOL FOR REQUEST

REQUEST URI	http://www.4glte.rw/voucher/api/external_access.jsp		
PROTOCOL TYPE	HTTP	METHOD	POST
PORT	80	CONTENTS TYPE	APPLICATION/XML
I/F CYCLE	EVERY REQUEST		

	TIME		

■ REQUEST ITEM / SCHEMA

SEND PARAMETERS (**NAME, VALUE**) : (REQ_XML, XML COMPRISING THE FOLLOWING ELEMENTS)

ELEMENT	M/O	FORMAT	DESCRIPTION
REQUEST	**M**		**REQUEST ROOT**
HEADER	**M**		**REQUEST HEADER**
SVCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
BODY	**M**		**REQUEST PARAMETER BODY**
SO_ID	M	STRING(4)	RETAILER ID
CUST_TP	M	STRING(2)	customer type A: Individual B: Self-Employee C: Foreigner D: Corporate E: Public Sector
CUST_NM	M	STRING(50)	Customer name
CORP_REG_NO	M	STRING(50)	Corporation number (CUST_TP: A,B,C,D Mandatory) If the customer type is 'E', send it the Business number. (ENCRYPTED WITH AES, 256BIT)

■ RESPONSE ITEM / SCHEMA

ELEMENT	M/O	FORMAT	DESCRIPTION
RESPONSE	**M**		**RESPONSE ROOT**
HEADER	**M**		**RESPONSE HEADER**
SVCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
MSGCODE	O	STRING(8)	MESSAGE CODE
MESSAGE	O	STRING(32)	MESSAGE DESCRIPTION
BODY	**M**		**RESPONSE PARAMETER BODY**
CUST_ID	M	STRING(10)	CUSTOMER ID

		CUST_NM	M	STRING(50)	CUSTOMER NAME

■ EXAMPLE :

YOU SHOULD SEND THE PARAMETER THAT IS SET TO THE STRING IS CONTAINING THE FOLLOWING REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<REQUEST>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
        <OPCODE>HPOP502</OPCODE>
        <CHCODE>A</CHCODE>
    </HEADER>
    <BODY>
        <SO_ID>02</SO_ID>
        <CUST_TP>A</CUST_TP>
        <CUST_NM>DAVID</CUST_NM>
        <CORP_REG_NO>01SD3152=2QWSDF</CORP_REG_NO>
    </BODY>
</REQUEST>
```

- **RESPONSE XML**

```
<RESPONSE>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
        <OPCODE>HPOP501</OPCODE>
        <CHCODE>A</CHCODE>
        <MSGCODE>1001</MSGCODE>
        <MESSAGE>…</MESSAGE>
    </HEADER>
    <BODY>
        <CUST_ID>C000111222</CUST_ID>
        <CUST_NM>DAVID</CUST_NM>
    </BODY>
</RESPONSE>
```

2.6. Searching MSISDN

2.6.1. Checking MSISDN status (SVCCODE: HP00005, OPCODE: HPOP601)

■ SUMMARY

YOU CAN BE SEARCHING MSISDN STATUS.

■ TELECOMMUNICATION PROTOCOL FOR REQUEST

REQUEST URI	http://www.4glte.rw/voucher/api/external_access.jsp		
PROTOCOL TYPE	HTTP	METHOD	POST
PORT	80	CONTENTS TYPE	APPLICATION/XML
I/F CYCLE	EVERY REQUEST TIME		

■ REQUEST ITEM / SCHEMA

SEND PARAMETERS (**NAME, VALUE**) : (REQ_XML, XML COMPRISING THE FOLLOWING ELEMENTS)

ELEMENT	M/O	FORMAT	DESCRIPTION
REQUEST	**M**		**REQUEST ROOT**
HEADER	**M**		**REQUEST HEADER**
SVCCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
BODY	**M**		**REQUEST PARAMETER BODY**
MSISDN	M	STRING(10)	MSISDN (CUSTMER PHONE NUMBER)
SO_ID	M	STRING(3)	RETAILER CODE

■ RESPONSE ITEM / SCHEMA

ELEMENT	M/O	FORMAT	DESCRIPTION
RESPONSE	**M**		**RESPONSE ROOT**
HEADER	**M**		**RESPONSE HEADER**
SVCCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
MSGCODE	O	STRING(8)	MESSAGE CODE
MESSAGE	O	STRING(32)	MESSAGE DESCRIPTION
BODY	**M**		**RESPONSE PARAMETER BODY**
RESULT	M	STRING(1)	Y : AVAILABLE N : NOT AVAILABLE

■ EXAMPLE :

YOU SHOULD SEND THE PARAMETER THAT IS SET TO THE STRING IS CONTAINING THE FOLLOWING REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<REQUEST>
 <HEADER>
   <SVCODE>HP00005</SVCODE>
   <OPCODE>HPOP601</OPCODE>
   <CHCODE>A</CHCODE>
 </HEADER>
 <BODY>
      <MSISDN>0771000050</MSISDN>
      <SO_ID>02</SO_ID>
</BODY>
</REQUEST>
```

- **RESPONSE XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<RESPONSE>
```

```
<HEADER>
    <SVCODE>HP00005</SVCODE>
    <OPCODE>HPOP601</OPCODE>
    <CHCODE>A</CHCODE>
    <MSGCODE>1001</MSGCODE>
    <MESSAGE>IT HAS BEEN SUCCESSFULLY PROCESSED.</MESSAGE>
</HEADER>
<BODY>
    <RESULT>Y</RESULT>
</BODY>
</RESPONSE>
```

2.6.2. Recommend 5 MSISDNS (SVCCODE: HP00005, OPCODE: HPOP602)

■ SUMMARY

YOU CAN GET RECOMMENDED 5 MSISDNS WITHIN SPECIFIC ISP.

■ TELECOMMUNICATION PROTOCOL FOR REQUEST

REQUEST URI	http://www.4glte.rw/voucher/api/external_access.jsp		
PROTOCOL TYPE	HTTP	METHOD	POST
PORT	80	CONTENTS TYPE	APPLICATION/XML
I/F CYCLE	EVERY REQUEST TIME		

■ REQUEST ITEM / SCHEMA

SEND PARAMETERS (**NAME, VALUE**) : (REQ_XML, XML COMPRISING THE FOLLOWING ELEMENTS)

ELEMENT	M/O	FORMAT	DESCRIPTION
REQUEST	**M**		**REQUEST ROOT**
HEADER	**M**		**REQUEST HEADER**
SVCCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
BODY	**M**		**REQUEST PARAMETER BODY**
SO_ID	M	STRING(3)	RETAILER ID

■ RESPONSE ITEM / SCHEMA

ELEMENT			M/O	FORMAT	DESCRIPTION
RESPONSE			**M**		**RESPONSE ROOT**
	HEADER		**M**		**RESPONSE HEADER**
		SVCCODE	M	STRING(8)	SERVICE CODE
		OPCODE	M	STRING(8)	OPERATION CODE
		CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
		MSGCODE	O	STRING(8)	MESSAGE CODE
		MESSAGE	O	STRING(32)	MESSAGE DESCRIPTION
	BODY		**M**		**RESPONSE PARAMETER BODY**
		MSISDN	M	STRING(512)	RECOMMENDED 5 MSISDN LIST

■ EXAMPLE :

YOU SHOULD SEND THE PARAMETER THAT IS SET TO THE STRING IS CONTAINING THE FOLLOWING REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<REQUEST>
 <HEADER>
   <SVCODE>HP00005</SVCODE>
   <OPCODE>HPOP602</OPCODE>
   <CHCODE>A</CHCODE>
 </HEADER>
 <BODY>
      <SO_ID>03</SO_ID>
</BODY>
</REQUEST>
```

- **RESPONSE XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<RESPONSE>
    <HEADER>
        <SVCODE>HP00005</SVCODE>
```

```
        <OPCODE>HPOP602</OPCODE>
        <CHCODE>A</CHCODE>
        <MSGCODE>XXXX</MSGCODE>
        <MESSAGE>....</MESSAGE>
    </HEADER>
    <BODY>
        <MSISDN >
            0770111111,0770111112,0770111113,0770111114,0770111115
        </MSISDN>
    </BODY>
</RESPONSE>
```

2.6.3. Searching MSISDN (SVCCODE: HP00001, OPCODE: HPOP003)

■ Summary

You can be searching MSISDN.

■ Telecommunication Protocol for Request

Request URI	http://www.4glte.rw/api_access.jsp		
Protocol Type	http	Method	POST
Port	80	Contents Type	Application/xml
I/F Cycle	Every Request time		

■ Request Item / Schema

Send parameters (**Name, Value**) : (REQ_XML, XML comprising the following elements)

Element	M/O	Format	Description
request	**M**		**Request Root**
header	**M**		**Request Header**
SVCCODE	M	String(7)	Service Code
OPCODE	M	String(7)	Operation Code
chcode	M	String(2)	Incoming Channel (default : A)
body	**M**		**Request Parameter Body**
ICCID	M	STRING(18)	ICCID
NATIONAL_ID	O	String(16)	National ID or TIN
ISP_CD	O	String(3)	Business No (defined by ORN)

■ Response Item / Schema

Element	M/O	Format	Description
response	**M**		**Response Root**
header	**M**		**Response Header**
SVCCODE	M	String(7)	Service Code
OPCODE	M	String(7)	Operation Code
chcode	M	String(2)	Incoming Channel (default : A)
msgcode	O	String(5)	Message Code
message	O	String(30)	Message Description
body	**M**		**Response Parameter Body**
SO_ID	M	STRING(3)	RETAILER CODE
SO_NM	M	STRING(100)	RETAILER NAME
MSISDN	M	String(15)	MSISDN

■ Example :

You should send the parameter that is set to the string is containing the following Request XML.(The Parameter's name is **REQ_XML**)

- Request XML

```
<?xml version='1.0' encoding='UTF-8' standalone='yes' ?>
<request>
 <header>
   <svcode>HP00001</svcode>
   <OPCODE>HPOP003</OPCODE>
   <CHCODE>A</CHCODE>
 </header>
 <body>
     <ICCID>9999999</ ICCID>
     <NATIONAL_ID></NATIONAL_ID>
     <ISP_CD>62</ISP_CD>
</body>
</request>
```

2.6.4. Searching Balance (SVCCODE: HP00005, OPCODE: HPOP101)

- SUMMARY

  YOU CAN BE SEARCHING BALANCE.

- TELECOMMUNICATION PROTOCOL FOR REQUEST

REQUEST URI	http://www.4glte.rw/voucher/api/external_access.jsp		
PROTOCOL TYPE	HTTP	METHOD	POST
PORT	80	CONTENTS TYPE	APPLICATION/XML
I/F CYCLE	EVERY REQUEST TIME		

- REQUEST ITEM / SCHEMA

  SEND PARAMETERS (**NAME, VALUE**) : (REQ_XML, XML COMPRISING THE FOLLOWING ELEMENTS)

ELEMENT	M/O	FORMAT	DESCRIPTION
REQUEST	**M**		**REQUEST ROOT**
HEADER	**M**		**REQUEST HEADER**
SVCCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
BODY	**M**		**REQUEST PARAMETER BODY**
MSISDN	M	STRING(10)	MSISDN (CUSTMER PHONE NUMBER)
SO_NM	M	STRING(100)	RETAILER NAME

■ RESPONSE ITEM / SCHEMA

ELEMENT	M/O	FORMAT	DESCRIPTION
RESPONSE	**M**		**RESPONSE ROOT**
HEADER	**M**		**RESPONSE HEADER**
SVCCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
MSGCODE	O	STRING(8)	MESSAGE CODE
MESSAGE	O	STRING(32)	MESSAGE DESCRIPTION
BODY	**M**		**RESPONSE PARAMETER BODY**
MSISDN	M	STRING(10)	MSISDN (CUSTMER PHONE NUMBER)
SO_ID	M	STRING(3)	RETAILER ID
REMAIN_AMT	M	STRING(128)	REMAIN 4G MONEY
TOTAL_VOLUME	M	STRING(128)	PURCHASED VOLUME
AMOUNT_USED	M	STRING(128)	TOTAL USAGE
TODAY_VOLUME	M	STRING(128)	TODAY USAGE
REMAIN_VOLUME	M	STRING(128)	REMAIN VOLUME

■ MESSAGE

MESSAGE CODE	MESSAGE DESCRIPTION
-8001	THERE IS NO MATCHING INFORMATION FOR MSISDN OR DISTINGUISH_ID.
-5002	MSISDN'S BUSINESS NO. AND TRANSFER XML'S BUSINESS NO. DO NOT MATCH.

■ EXAMPLE :

YOU SHOULD SEND THE PARAMETER THAT IS SET TO THE STRING IS CONTAINING THE FOLLOWING REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML**

<?XML VERSION="1.0" ENCODING="UTF-8"?>

<REQUEST>

```
<HEADER>
  <SVCODE>HP00005</SVCODE>
  <OPCODE>HPOP101</OPCODE>
  <CHCODE>A</CHCODE>
</HEADER>
<BODY>
    <MSISDN>0771000050</MSISDN>
</BODY>
</REQUEST>
```

- **RESPONSE XML**

```
<?XML VERSION="1.0" ENCODING="UTF-8"?>
<RESPONSE>
  <HEADER>
    <SVCODE>HP00005</SVCODE>
    <OPCODE>HPOP101</OPCODE>
    <CHCODE>A</CHCODE>
    <MSGCODE>1001</MSGCODE>
    <MESSAGE>IT HAS BEEN SUCCESSFULLY PROCESSED.</MESSAGE>
  </HEADER>
  <BODY>
    <MSISDN>0771000050</MSISDN>
    <REMAIN_AMT>23000</REMAIN_AMT>
    <TOTAL_VOLUME>5000</TOTAL_VOLUME>
    <AMOUNT_USED>2200</AMOUNT_USED>
    <TODAY_VOLUME>500</TODAY_VOLUME>
    <REMAIN_VOLUME>2800</REMAIN_VOLUME>
  </BODY>
</RESPONSE>
```

2.6.5. Searching Currency Balance (SVCCODE: HP00001, OPCODE: HPOP001)

■ Summary

You can be searching currency balance.

■ Telecommunication Protocol for Request

Request URI	http://www.4glte.rw/api_access.jsp		
Protocol Type	http	Method	POST
Port	80	Contents Type	Application/xml
I/F Cycle	Every Request time		

■ Request Item / Schema

Send parameters (**Name, Value**) : (REQ_XML, XML comprising the following elements)

Element	M/O	Format	Description
request	**M**		**Request Root**
header	**M**		**Request Header**
SVCCODE	M	String(7)	Service Code
OPCODE	M	String(7)	Operation Code
chcode	M	String(2)	Incoming Channel (default : A)
body	**M**		**Request Parameter Body**
MSISDN	M	String(15)	MSISDN
ISP_CD	M	String(3)	Business No (defined by ORN)

■ Response Item / Schema

Element			M/O	Format	Description
response			**M**		**Response Root**
	header		**M**		**Response Header**
		SVCCODE	M	String(7)	Service Code
		OPCODE	M	String(7)	Operation Code
		chcode	M	String(2)	Incoming Channel (default : A)
		msgcode	O	String(5)	Message Code
		message	O	String(30)	Message Description
	body		**M**		**Response Parameter Body**
		REMAIN_AMT	M	String(15)	Remaining Money

■ Example :

You should send the parameter that is set to the string is containing the following Request XML.(The Parameter's name is **REQ_XML**)

- Request XML

```
<?xml version='1.0' encoding='UTF-8' standalone='yes' ?>
<request>
 <header>
   <svcode>HP00001</svcode>
   <OPCODE>HP0P001</OPCODE>
   <CHCODE>A</CHCODE>
 </header>
 <body>
     <MSISDN>0771000050</MSISDN>
     <ISP_CD>62</ISP_CD>
</body>
</request>
```

2.6.6. Balance Transfer (SVCCODE: HP00002, OPCODE: HPOP005)

■ SUMMARY

YOU CAN TRANSFER YOUR BALANCE FROM ONE SUBSCRIBER TO ANOTHER SUBSCRIBER.

■ TELECOMMUNICATION PROTOCOL FOR REQUEST

REQUEST URI	http://www.4glte.rw/voucher/api/external_access.jsp		
PROTOCOL TYPE	HTTP	METHOD	POST
PORT	80	CONTENTS TYPE	APPLICATION/XML
I/F CYCLE	EVERY REQUEST TIME		

■ REQUEST ITEM / SCHEMA

SEND PARAMETERS (**NAME, VALUE**) : (REQ_XML, XML COMPRISING THE FOLLOWING ELEMENTS)

ELEMENT	M/O	FORMAT	DESCRIPTION
REQUEST	**M**		**REQUEST ROOT**
HEADER	**M**		**REQUEST HEADER**
SVCCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
BODY	**M**		**REQUEST PARAMETER BODY**
SRC_MSISDN	M	STRING(10)	SOURCE MSISDN (CUSTMER PHONE NUMBER)
ISP_CD	M	STRING(3)	BUSINESS NO.
REQ_SYS_ID	M	STRING(2)	DISTINGUISHED REQUEST TYPE
SUBS_PWD	M	STRING(20)	SUBSCRIBER PASSWORD OF LOGIN ECARE (ENCRYPTED WITH AES, 256BIT) KEY: aes256-ktrn-key1
BAL_TYPE	M	STRING(5)	BALANCE TYPE ID
TRANS_AMT	M	STRING	AMOUNT OF TRANSFER BALANCE, IT CAN BE MONEY OR VOLUME BALANCE

		DEST_MSISDN	M	STRING(10)	DESTINATION MSISDN (CUSTMER PHONE NUMBER)

■ RESPONSE ITEM / SCHEMA

ELEMENT	M/O	FORMAT	DESCRIPTION
RESPONSE	**M**		**RESPONSE ROOT**
HEADER	**M**		**RESPONSE HEADER**
SVCCODE	M	STRING(8)	SERVICE CODE
OPCODE	M	STRING(8)	OPERATION CODE
CHCODE	M	STRING(2)	INCOMING CHANNEL (DEFAULT : A)
MSGCODE	O	STRING(8)	MESSAGE CODE
MESSAGE	O	STRING(32)	MESSAGE DESCRIPTION

■ MESSAGE

MESSAGE CODE	MESSAGE DESCRIPTION
-8001	THERE IS NO MATCHING INFORMATION FOR MSISDN OR DISTINGUISH_ID.
-7100	THE BALANCE OF TRANSFER IS DENIED
-7101	BEYOND MAXIMUM BALANCE LIMITATION OF SOURCE MSISDN
-7102	BIGGER THAN MINIMUM BALANCE LIMITATION OF TARGET MSISDN
-7103	PLEASE CHANGE DEFAULT PASSWORD AS NEW ONE, THEN REQUEST BALANCE TRANSFERATION

■ EXAMPLE :

YOU SHOULD SEND THE PARAMETER THAT IS SET TO THE STRING IS CONTAINING THE FOLLOWING REQUEST XML.(THE PARAMETER'S NAME IS **REQ_XML**)

- **REQUEST XML**

```
<REQUEST>
  <HEADER>
    <SVCCODE>HP00002</SVCCODE>
    <OPCODE>HPOP005</OPCODE>
    <CHCODE>A</CHCODE>
```

```
    </HEADER>
    <BODY>
      <ISP_CD>56</ISP_CD>
      <SRC_MSISDN>250788561340</SRC_MSISDN>
      <REQ_SYS_ID>01</REQ_SYS_ID>
      <SUBS_PWD>000000</SUBS_PWD>
      <BAL_TYPE>1</BAL_TYPE>
      <TRANS_AMT>1300</TRANS_AMT>
      <DEST_MSISDN>250788561373</DEST_MSISDN>
    </BODY>
  </REQUEST>
```

- **RESPONSE XML**

```
<?XML VERSION="1.0" ENCODING="UTF-
8" STANDALONE="NO"?><RESPONSE><HEADER><MSGCODE>1001</MSGCODE>< S V C C O D E
>HP00002</ S V C C O D E ><CHCODE>A</CHCODE>< O P C O D E >HPOP005</ O P C O D E
></HEADER></RESPONSE>
```

2.6.7. Request for recharge (SVCCODE: HP00002, OPCODE: HPOP001)

■ Summary

You can make a **recharge request**.

■ Telecommunication Protocol for Request

Request URI	http://www.4glte.rw/api_access.jsp		
Protocol Type	http	Method	POST
Port	80	Contents Type	Application/xml
I/F Cycle	Every Request time		

■ Request Item / Schema

Send parameters (**Name, Value**) : (REQ_XML, XML comprising the following elements)

Element	M/O	Format	Description
request	**M**		**Request Root**
header	**M**		**Request Header**
SVCCODE	M	String(7)	Service Code

OPCODE	M	String(7)	Operation Code
chcode	M	String(2)	Incoming Channel (default : A)
body	**M**		**Request Parameter Body**
MSISDN	M	String(15)	MSISDN
NATIONAL_ID	M	String(16)	National ID or TIN
ISP_CD	M	String(3)	Business No (defined by ORN)
REQ_SYS_ID	M	String(2)	Distinguished Request Type 01 : Pivot Access
RCHRG_TP	M	String	Recharge Type 100 : Recharge Data
RCHRG_STAT	M	String	Recharge Status 10 : Recharge

		RCHRG_APPR_NO	M	String	Pivot Access's Approval No
		RCHRG_AMT	M	String	Recharge Money

■ Response Item / Schema

Element			M/O	Format	Description
response			**M**		**Response Root**
	header		**M**		**Response Header**
		SVCCODE	M	String(7)	Service Code
		OPCODE	M	String(7)	Operation Code
		chcode	M	String(2)	Incoming Channel (default : A)
		msgcode	O	String(5)	Message Code
		message	O	String(30)	Message Description

■ Example :

You should send the parameter that is set to the string is containing the following Request XML.(The Parameter's name is **REQ_XML**)

- Request XML

```
<?xml version='1.0' encoding='UTF-8' standalone='yes' ?>
<request>
 <header>
   <svcode>HP00002</svcode>
   <OPCODE>HP0P001</OPCODE>
   <CHCODE>A</CHCODE>
 </header>
 <body>
     <MSISDN>0771000050</MSISDN>
     <NATIONAL_ID>8012101234567890</NATIONAL_ID>
     <ISP_CD>62</ISP_CD>
     <REQ_SYS_ID>01</REQ_SYS_ID>
     <RCHRG_TP>100</RCHRG_TP>
     <RCHRG_STAT>10</RCHRG_STAT>
     <RCHRG_APPR_NO>9999999999</RCHRG_APPR_NO>
     <RCHRG_AMT>10000</RCHRG_AMT>
```

</body>

</request>

2.6.8. Cancel for recharging (SVCCODE: HP00002, OPCODE: HPOP002)

■ Summary

You can make a recharge cancel..

■ Telecommunication Protocol for Request

Request URI	http://www.4glte.rw/api_access.jsp		
Protocol Type	http	Method	POST
Port	80	Contents Type	Application/xml
I/F Cycle	Every Request time		

■ Request Item / Schema

Send parameters (Name, Value) : (REQ_XML, XML comprising the following elements)

Element	M/O	Format	Description
request	**M**		**Request Root**
header	**M**		**Request Header**
SVCCODE	M	String(7)	Service Code
OPCODE	M	String(7)	Operation Code
chcode	M	String(2)	Incoming Channel (default : A)
body	**M**		**Request Parameter Body**
MSISDN	M	String(15)	MSISDN
NATIONAL_ID	M	String(16)	National ID or TIN
ISP_CD	M	String(3)	Business No (defined by ORN)
REQ_SYS_ID	M	String(2)	Distinguished Request Type 01 : Pivot Access
RCHRG_TP	M	String	Recharge Type 100 : Recharge Data
RCHRG_STAT	M	String	Recharge Status 30 : Cancel Recharge
RCHRG_APPR_NO	M	String	Pivot Access's Approval No
CNL_AMT	M	String	Cancel money

■ Response Item / Schema

Element	M/O	Format	Description
response	**M**		**Response Root**
header	**M**		**Response Header**
SVCCODE	M	String(7)	Service Code
OPCODE	M	String(7)	Operation Code
chcode	M	String(2)	Incoming Channel (default : A)
msgcode	O	String(5)	Message Code
message	O	String(30)	Message

■ Example :

You should send the parameter that is set to the string is containing the following Request XML.(The Parameter's name is **REQ_XML**)

- Request XML

```
<?xml version='1.0' encoding='UTF-8' standalone='yes' ?>
<request>
 <header>
   <svcode>HP00001</svcode>
   <OPCODE>HP0P001</OPCODE>
   <CHCODE>A</CHCODE>
 </header>
 <body>
     <MSISDN>0771000050</MSISDN>
     <NATIONAL_ID>8012101234567890</NATIONAL_ID>
     <ISP_CD>62</ISP_CD>
     <REQ_SYS_ID>01</REQ_SYS_ID>
     <RCHRG_TP>100</RCHRG_TP>
```

```
        <RCHRG_STAT>30</RCHRG_STAT>
        <RCHRG_APPR_NO>9999999999</RCHRG_APPR_NO>
        <CNL_AMT>10000</CNL_AMT>
    </body>
    </request>
```

2.7. KTRN Simreg API workflow (SVCCODE: HP00005)

Summary

This section describes the standard API interaction flow between the 3rd Party system and KTRN BSS for staff authentication, customer registration, and new connection activation. The 3rd Party system is responsible for performing the required business-rule validation based on the information returned by KTRN APIs before submitting the final transaction request.

2.7.1. staff login (API 201)

Before performing any SIM registration operation, the 3rd Party system shall call API 201 to retrieve the staff's authorized working location information.

1. The 3rd Party system shall call **API 201 - Request Auth Login (HPOP201)** to retrieve the staff-related information from KTRN BSS.
2. KTRN BSS will return the staff's authorized working location information, including:
    - Authorized longitude
    - Authorized latitude
    - Allowed working distance/radius (DISTANCE)
    - Organization and retailer information
3. The 3rd Party system shall store the LONGITUDE, LATITUDE, and DISTANCE returned by API 201 and use them for geographic validation during subsequent customer registration and new connection operations.
4. Before performing operations that require location validation, the 3rd Party system shall compare the staff's actual current GPS

location against the authorized location and working radius returned by API 201.

5. If the staff's current location is within the authorized working area, the related operation may continue. If the staff's current location is outside the authorized working area, the 3rd Party system shall reject the operation.

2.7.2. create customer (API 312 + API 501)

When registering a new customer, the 3rd Party system shall follow the workflow below.

a. Fingerprint Verification

During customer information collection, the customer's fingerprint must be verified before customer registration can continue.

1. The 3rd Party system shall collect the required customer identity and fingerprint information.
2. The 3rd Party system shall call API 312 – Verifying Fingerprint (HPOP312).
3. KTRN BSS will return the fingerprint verification result.
4. The 3rd Party system shall process the result as follows:
    - If the verification result is successful, the customer registration process may continue.
    - If the verification result fails, the 3rd Party system shall reject the operation and shall not proceed with customer registration.

API 312 returns S for successful fingerprint verification and F for failed verification.

b. Customer Registration

After all required customer information has been collected and the fingerprint has been successfully verified, the 3rd Party system shall perform the following steps before calling the customer registration API.

1. Validate the staff's current location

    The 3rd Party system shall obtain the staff's current GPS location and compare it with the authorized geographic information returned by API 201 during staff login.

    The 3rd Party system is responsible for performing this geographic validation.

- If the staff's current location is within the authorized working area, the operation may continue.
- If the staff's current location is outside the authorized working area, the 3rd Party system shall reject the operation and shall not submit the customer registration request to KTRN BSS.

2. Call API 501 – Subscribe Customer Info (HPOP501)

   After the location validation is successful, the 3rd Party system shall call API 501 to create the customer.

3. The API 501 request shall include:
   - Customer registration information
   - Staff's actual current longitude
   - Staff's actual current latitude

   The longitude and latitude sent to API 501 shall represent the **actual location where the registration operation is performed**, rather than the configured authorized location.

4. If API 501 is successfully processed, the 3rd Party system may continue with the subsequent SIM registration or new connection process.

API 501 is used for new customer registration and currently contains customer, organization, retailer, longitude, and latitude information in the interface specification.

2.7.3. New connection (API 313 + API 310 + API 312 + API 311 + API 302)

When activating a new SIM connection for a customer, the 3rd Party system shall perform subscriber-limit validation, fingerprint verification, OTP processing where applicable, and staff location validation before submitting the activation request.

a. Check Existing Subscriber Count

Before creating a new connection, the 3rd Party system shall determine how many active subscriptions are already registered under the customer's identity.

1. The 3rd Party system shall call API 313 – Encrypt/Decrypt Sensitive PWD/CERT (HPOP313) to encrypt the customer's National ID or other applicable identification number.

2. The encrypted identification number shall then be used to call API 310 – Searching Subscriber Count (HPOP310).
3. API 310 returns:
    - The number of existing subscriptions under the customer identity
    - The customer's first number, if applicable

    API 310 provides SUBSCOUNT and FISRTNUMBER in its response.

4. The 3rd Party system shall apply the subscriber-limit rules based on the customer type:
    - Rwandan customer: Maximum 3 subscriptions
    - Foreigner customer: Maximum 1 subscription
5. The subscriber-limit validation shall be performed by the 3rd Party system.

    If the customer has already reached the maximum allowed number of subscriptions, the 3rd Party system shall reject the new connection request and shall not proceed to activation.

b. Fingerprint Verification

Before the new connection can be activated, the customer fingerprint shall be verified.

1. The 3rd Party system shall call API 312 – Verifying Fingerprint (HPOP312).
2. The workflow shall continue only when the fingerprint verification is successful.
3. If fingerprint verification fails, the 3rd Party system shall reject the new connection operation.

c. OTP Verification for Additional SIM Registration

For a Rwandan customer who already has an existing subscription, an OTP shall be sent to the customer's first registered number before the additional SIM can be activated.

1. This step is required when:
    - The customer is a Rwandan customer; and
    - API 310 indicates that the customer already has at least one subscription; and
    - API 310 returns the customer's first registered number.
2. The 3rd Party system shall call API 311 – Get OTP Code (HPOP311).
3. The RNUMBER parameter shall contain the customer's first registered number returned by API 310.
4. KTRN BSS will trigger an OTP and send it to the specified first registered number.
5. The 3rd Party system shall collect the OTP entered by the customer and retain it for submission in the final activation request.

API 311 is specifically used to trigger an OTP to the customer's first number.

If the customer does not already have an existing subscription, this OTP step is not required.

d. Submit New Connection Request

After all required validations and customer information collection have been completed, the 3rd Party system shall submit the activation request using **API 302 – Request for Simple Activation (HPOP302)**.

Before calling API 302, the following validation shall be performed:

1. **Validate the staff's current location**

   The 3rd Party system shall compare the staff's actual current GPS location with the authorized working location information returned by API 201.

   This validation shall be performed by the 3rd Party system.

   - If the staff is within the authorized working area, the activation may continue.
   - If the staff is outside the authorized working area, the 3rd Party system shall reject the operation and shall not call API 302.

2. **Send the staff's actual operation location**

   The 3rd Party system shall populate the following fields in API 302 with the staff's actual current geographic location:

   - LONGITUDE
   - LATITUDE

   These values shall represent the actual location where the SIM registration operation is performed. API 302 supports longitude and latitude as part of the activation request.

3. **Send OTP information when applicable**

   If the OTP process described in Section 2.7.3(c) is required, the 3rd Party system shall include:

   - RNUMBER – the customer's first active number
   - OTP – the OTP entered by the customer

   API 302 provides both RNUMBER and OTP fields for this purpose.

4. Send customer verification attachments

   The 3rd Party system shall include the required customer verification information and attachments in the API 302 request, including:

   - Customer photo
   - Fingerprint data
   - Finger/hand information, where applicable
   - Other required registration information

   The current API 302 specification supports HAND, FINGERPRINT, and PHOTO information.

5. After all required information has been populated, the 3rd Party system shall call API 302.
6. If API 302 returns a successful result, the new connection process is completed. If the request fails, the 3rd Party system shall handle the failure according to the returned message code and shall not treat the SIM registration as successfully completed.



SimReg_API Changes Syntax.xlsx